# [BUSINESS] FORECASTING AND PREDICTIVE MODELING

## LECTURE 3

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# In Lecture 2 ...

- Data in R
- Visualization in R

# Today

- Time series visualization

# Time series visualizations

- The first thing to do when working with a time series is to plot it. We can use the function *autoplot* for this.

```
olympic_running |> autoplot()

olympic_running |>
  filter(Length == 100 & Sex == "men") |>
  autoplot()
```

# Time series plot

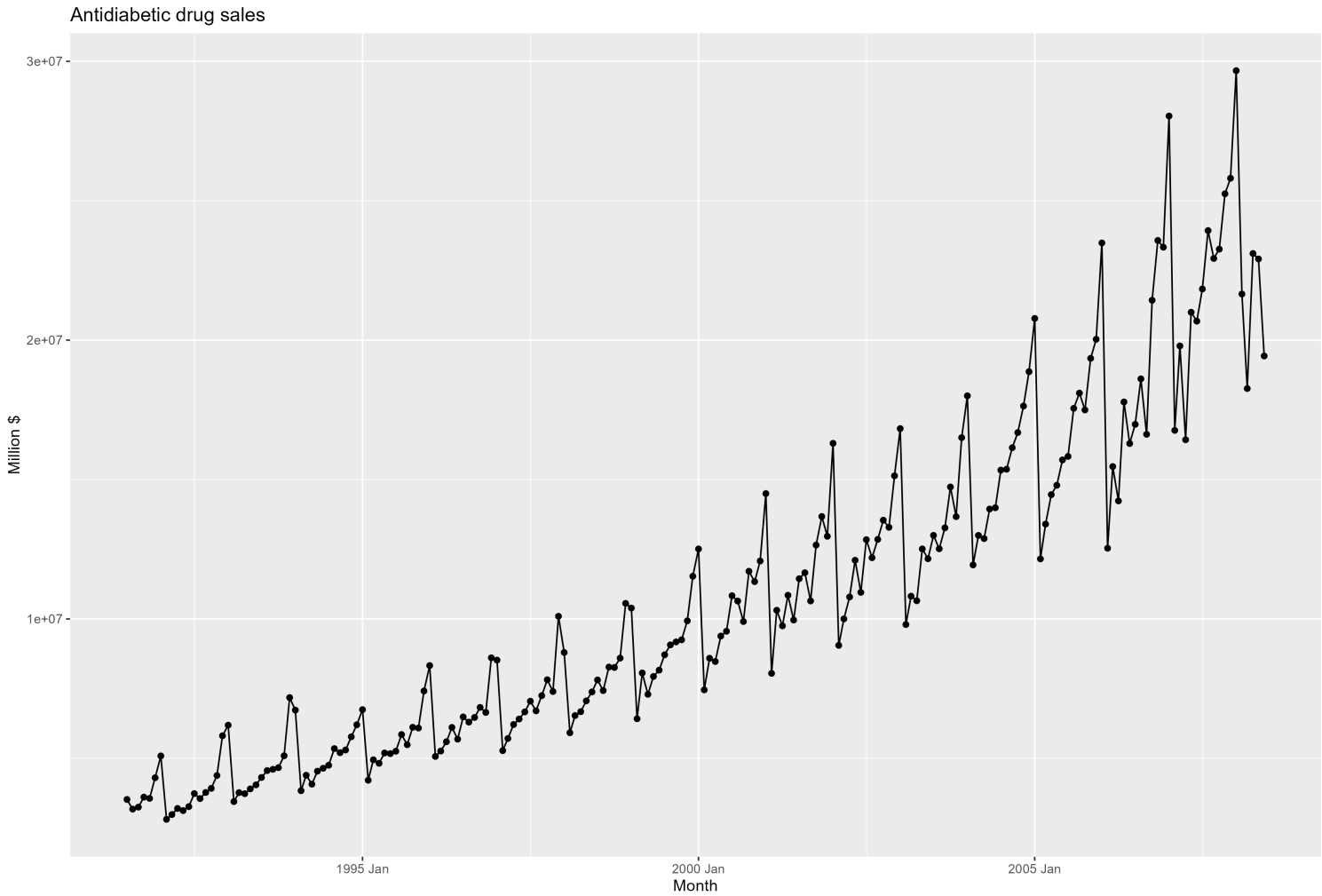
- Let us start with a data set called *PBS*, containing monthly drug prescription data from Australia from 1991 July to 2008 June. The drugs are divided into categories. We will only use a part of this data

```
a10 <- PBS |>
  filter(ATC2 == "A10") |>
  select(Month, Concession, Type, Cost) |>
  summarize(TotalCost = sum(Cost)) |>
  mutate(Cost = TotalCost / 1000000)

a10 |> autoplot(TotalCost) +
  labs(x = "Month", y = "Million $", title = "Antidiabetic drug sales")
```

- Similarly, we can create the following tsibbles:
  - A11 - Vitamins
  - A12 - Mineral supplements
  - J07 - Vaccines
  - NO2 - Analgesics

# Time series plot

- Sometimes, it is hard to identify the exact values on the time series plot. We can also solve this problem by showing the actual points, adding a *geom_point* layer to the plot.


Antidiabetic drug sales
3e+07
2e+07
1e+07
Million $
1995 Jan
2000 Jan
2005 Jan
Month

# Trend, season, cycle

- When visually analyzing a time series plot, there are three important terms that can be used to describe it.
  - **trend** - describes a long term increase or decrease in the data, even if it is not linear.
  - **seasonal pattern** - appears when the data values are affected by factors related to the calendar like time of the year or day of the week. Seasonality always has a fixed known frequency, which can be at most a year.
  - **cycle** - when there are rises and falls, but not of fixed frequency, caused in general by economic conditions. Their duration is usually at least 2 years.

- There is a fourth term that can be used to characterize time series data: **stationarity**.

- A time series is stationary if its statistical properties (mean, standard deviation) do not depend on the time at which the series is observed. This means that the time series cannot have trend or seasonality.

- Whether a time series is stationary or not, can be inspected visually, but there are also statistical tests that can be used to check this (these will be discussed later).

- Does the antidiabetic drug sales data have a trend?

- Does the antidiabetic drug sales data have a trend?
- Yes, it has an increasing trend.

- Does the antidiabetic drug sales data have seasons?

- Does the antidiabetic drug sales data have a trend?
- Yes, it has an increasing trend.

- Does the antidiabetic drug sales data have seasons?
- Yes, it seems to have seasonality.

- Does the antidiabetic drug sales data have cycles?

- Does the antidiabetic drug sales data have a trend?
- Yes, it has an increasing trend.

- Does the antidiabetic drug sales data have seasons?
- Yes, it seems to have seasonality.

- Does the antidiabetic drug sales data have cycles?
- No, it does not.

- In case of seasonal data, determining the length of a season is very important (it is a value which is used later in analysis).
- Another important thing to determine in case of seasonal data is whether the magnitude is constant over time (we call this additive effect) or whether it grows or shrinks (multiplicative effect).
- When you have many points on your plot (we have 204), determining the exact length might not be easy. In such cases we can *zoom in* on the data, by plotting only parts of it. We can use the *head, tail* or *slice* functions.

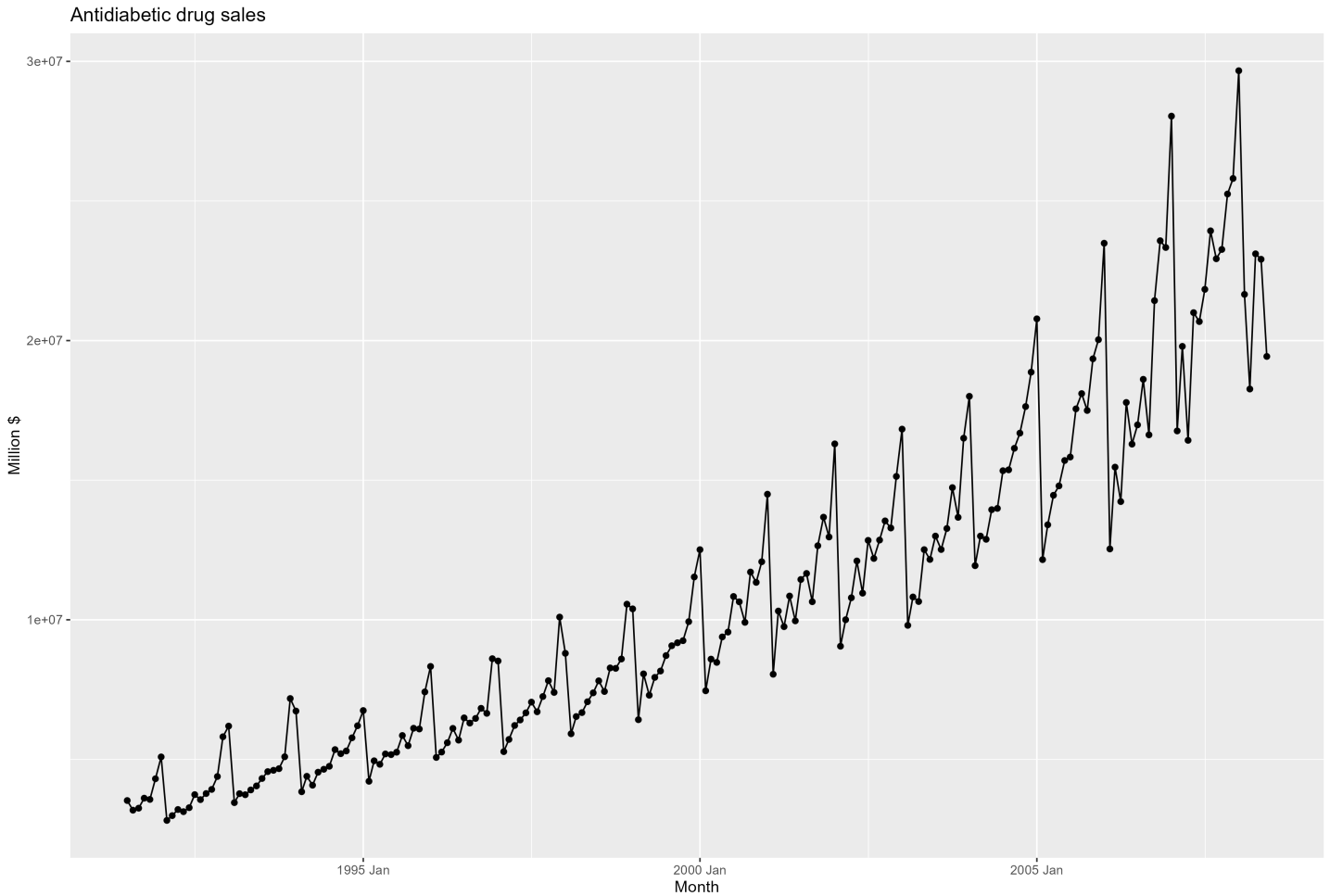
```
a10 |>
  head(n = 35) |>
  autoplot() +
  geom_point() +
 labs(x = "Month", y = "Million $", title = "First 35 observations for
     antidiabetic drug sales")
```

# Trend, season, cycle examples

- Let us see a few examples of time series from the PBS data.

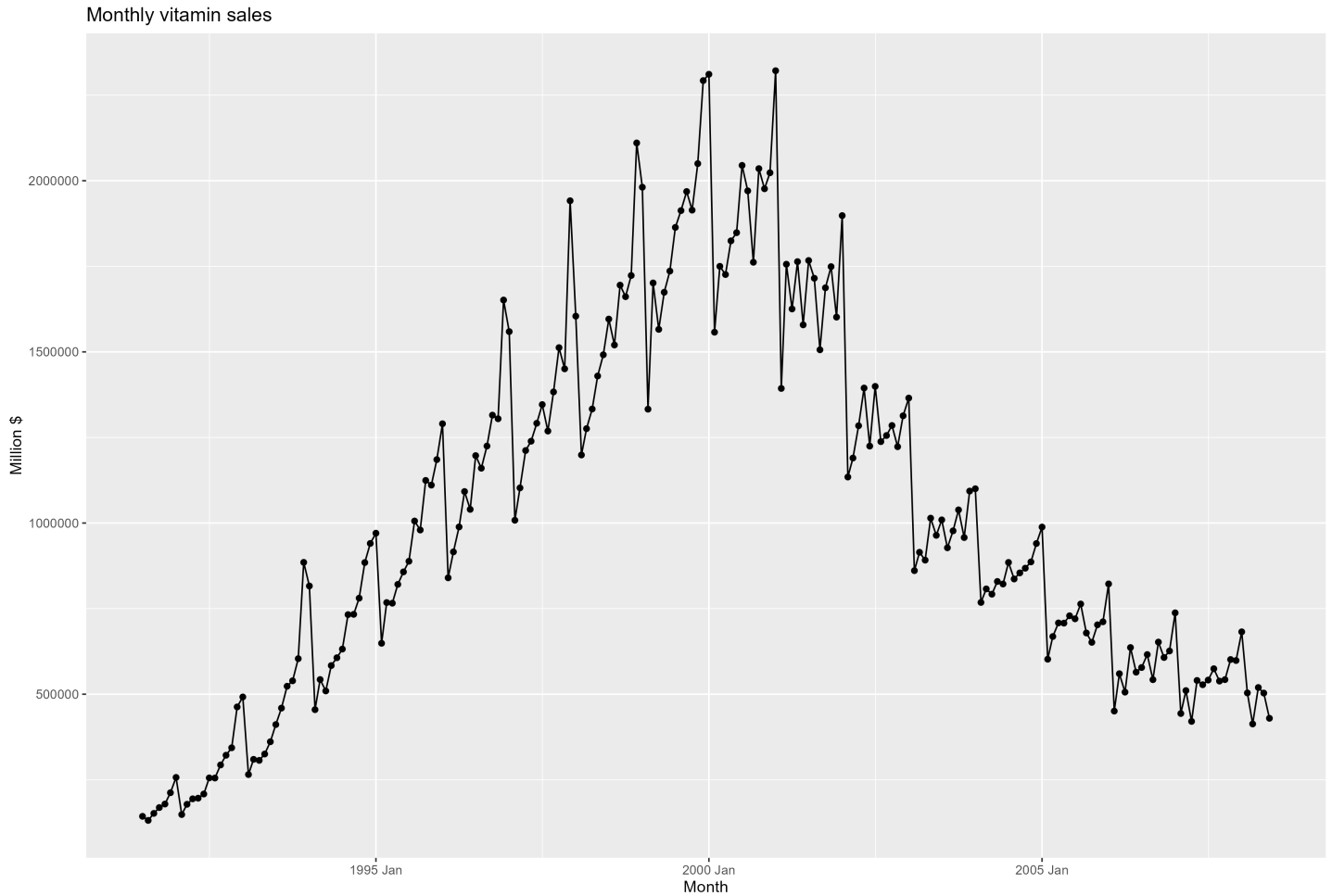
```
A11 |> autoplot(TotalCost) +
  geom_point() +
  labs(x = "Month", y = "Million $", title = "Monthly vitamin sales")

A12 |> autoplot(TotalCost) +
  geom_point() +
  labs(x = "Month", y = "Million $", title = "Monthly mineral supplements sales"
      )

N02|> autoplot(TotalCost) +
  geom_point() +
  labs(x = "Month", y = "Million $", title = "Monthly analgesics sales")
```

# Trend, season, cycle example

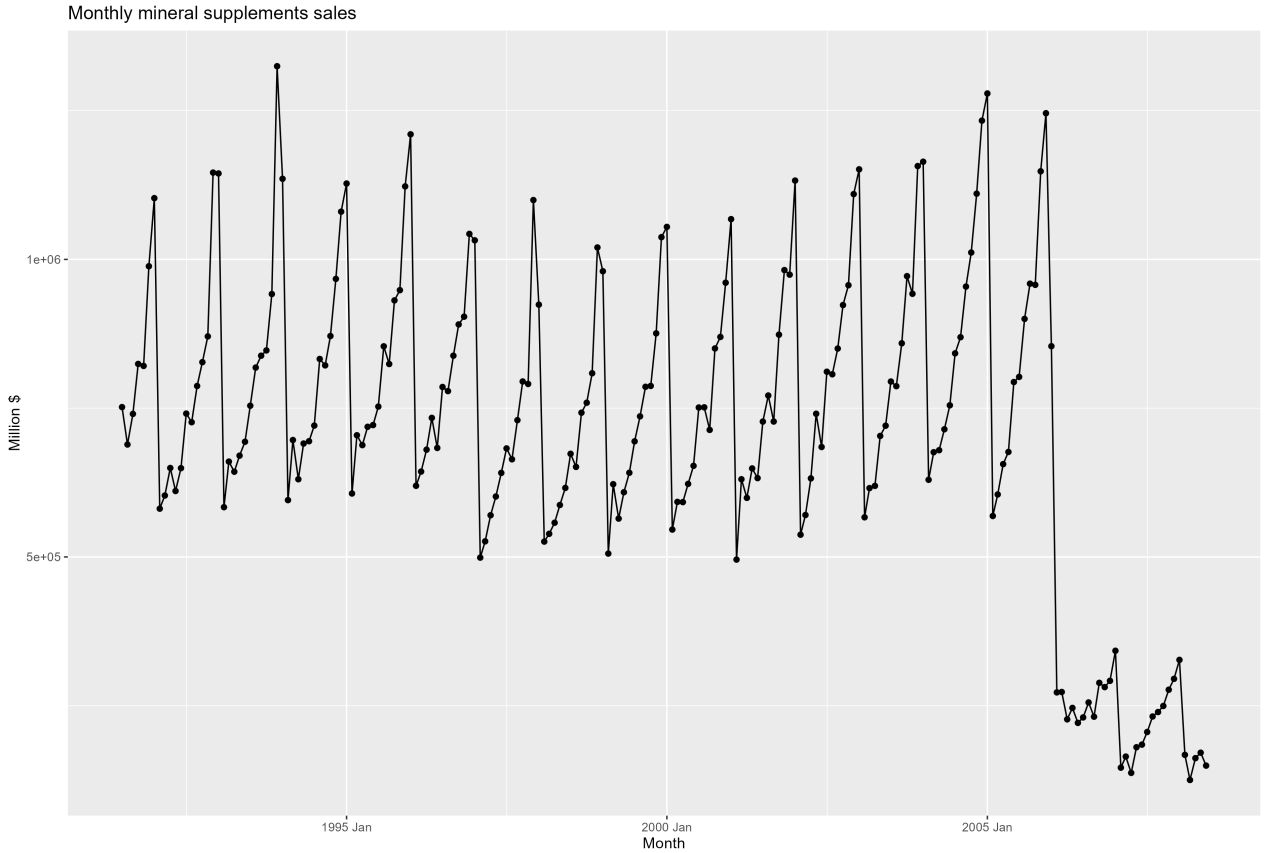



- Increasing trend, multiplicative seasonality

# Trend, season, cycle example

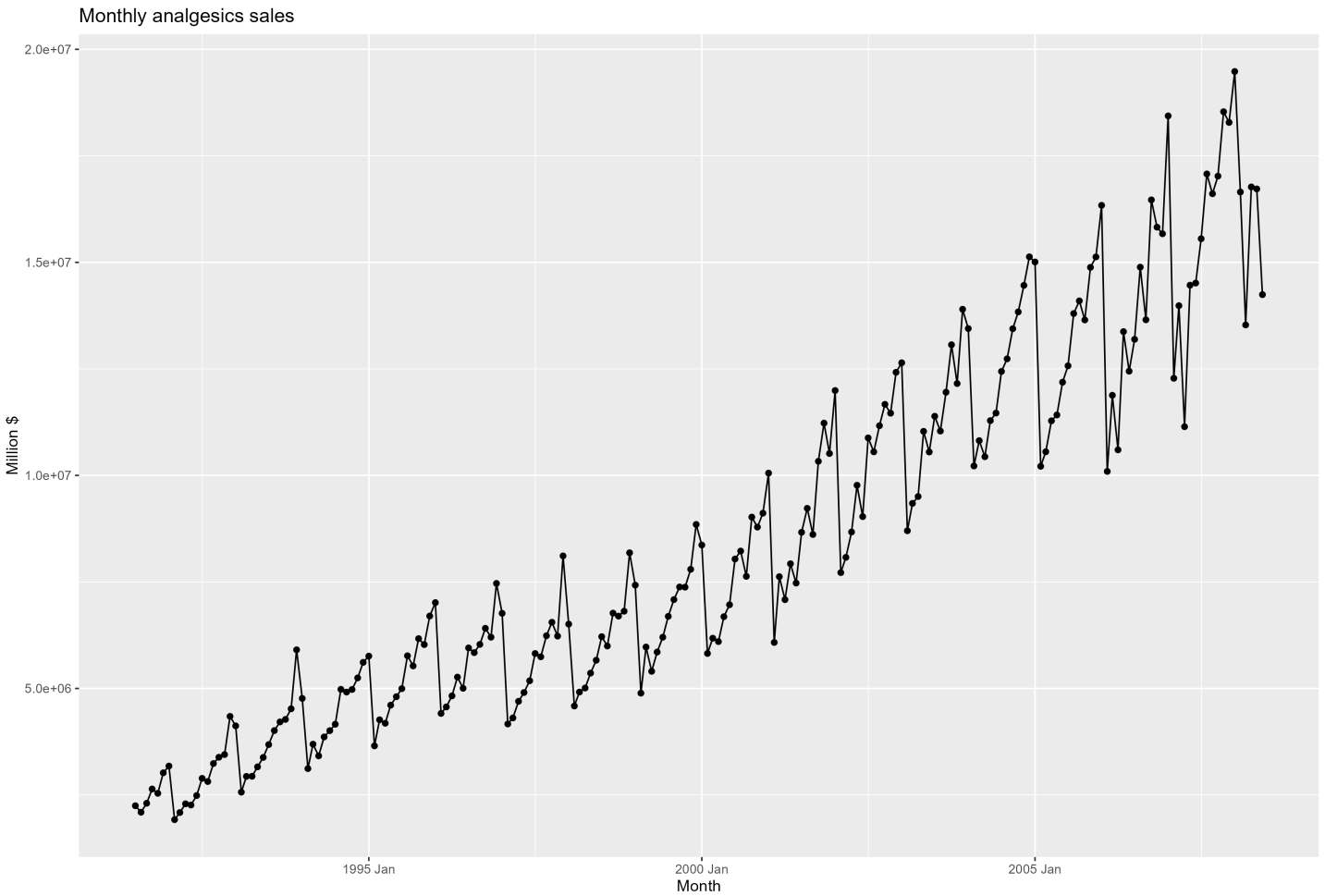



- Increasing trend until 2000, then a decreasing trend.
- Clear seasonality (not easy to determine if it is additive or multiplicative).




- Sudden drop in values in 2006 February.
- Until 2006 February there does not seem to be a trend (or only a very small one) and there is not enough data after 2006 to determine if there is a trend.
- There is seasonality, it seems to be additive (but not enough data to clearly see it after 2006)

# Trend, season, cycle example

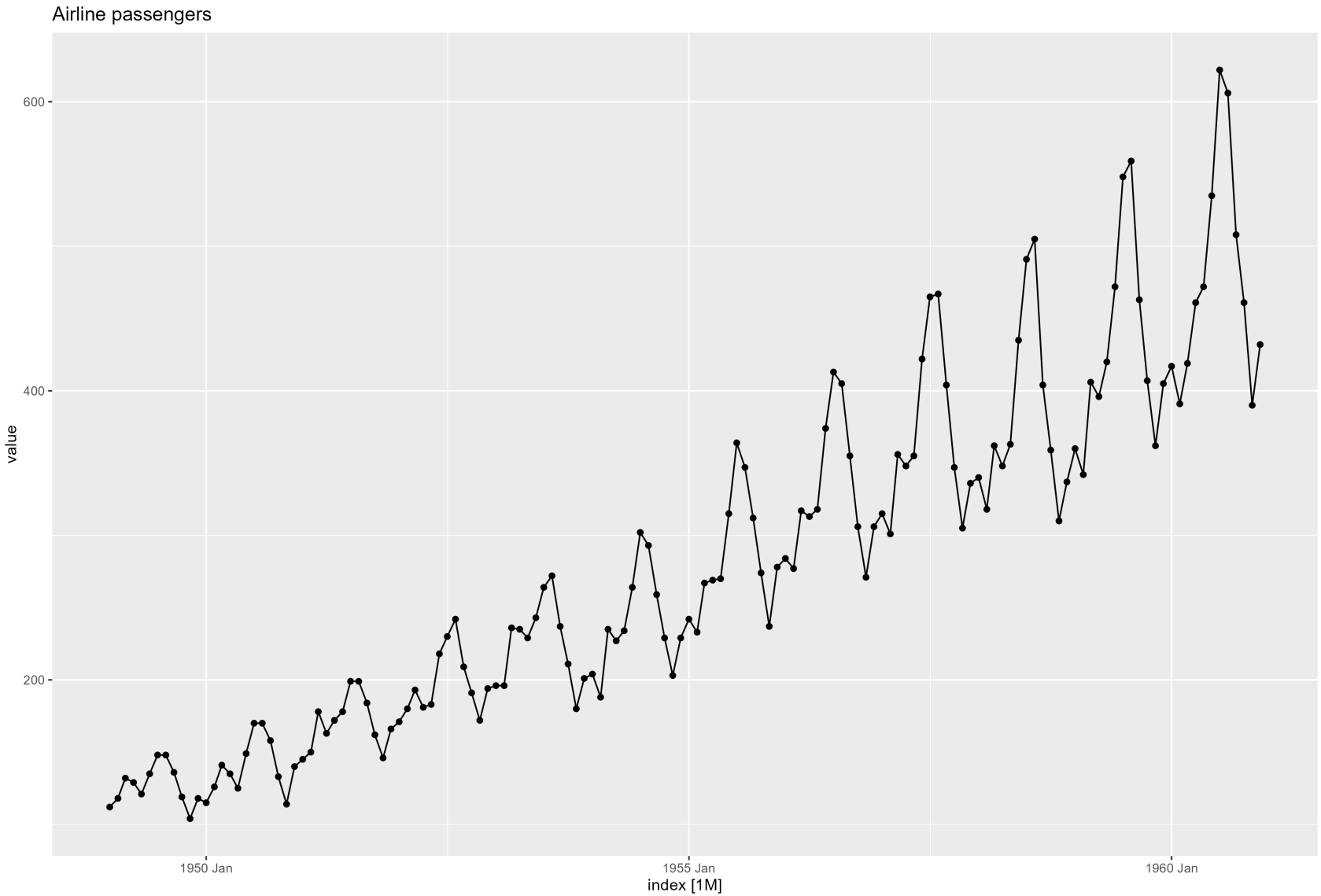



- Clearly increasing trend and multiplicative seasonality (very similar to antidiabetic drugs)

# Trend, season, cycle example

- Monthly total data about airline passengers from 1949 to 1960, in thousands (data from datasets package in R)

```
AirPassengers |>
  as_tsibble() |>
  autoplot() +
  geom_point() +
  labs(title = "Airline passengers")
```

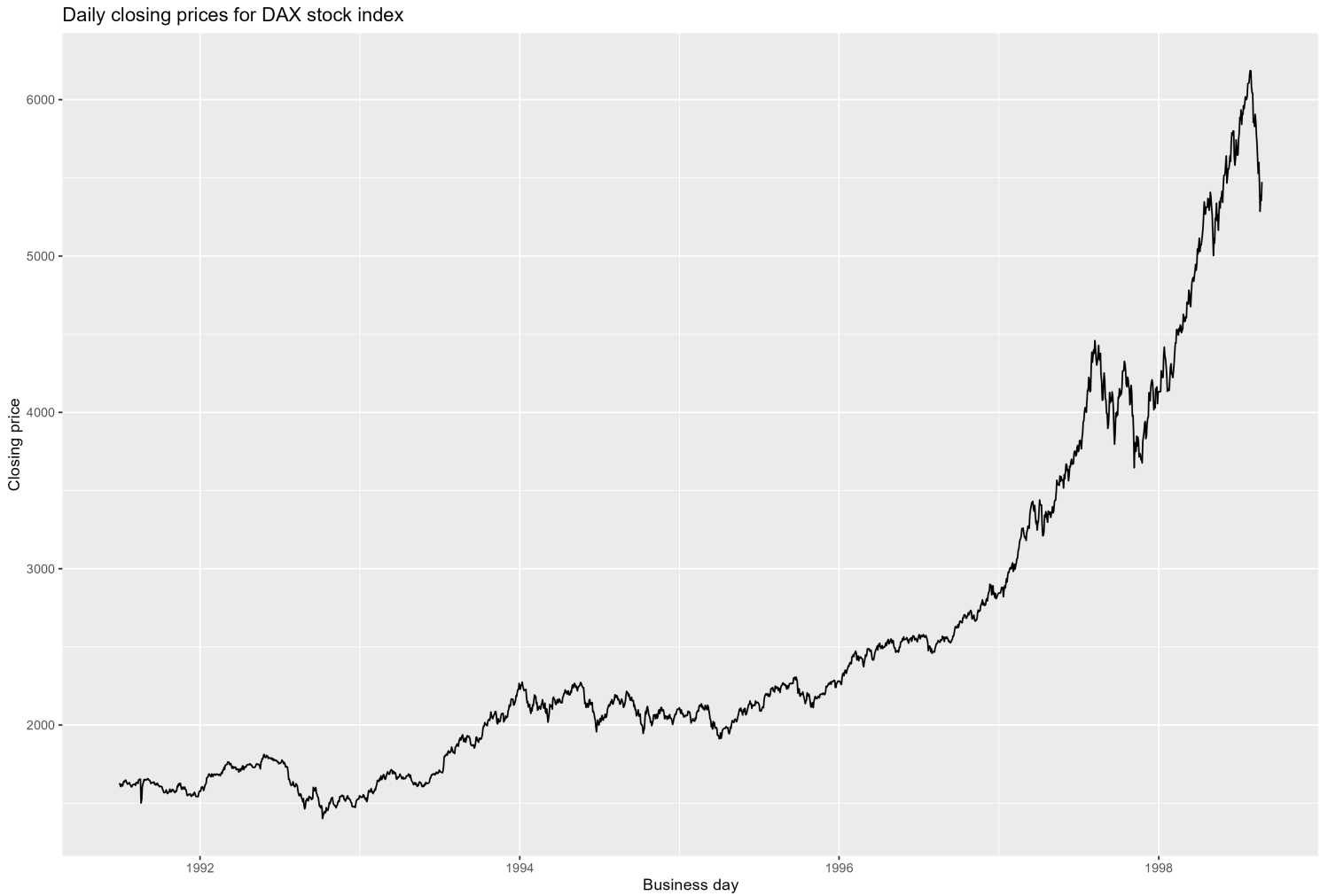



- Increasing trend and multiplicative seasonality

# Trend, season, cycle example

- Daily closing prices of the DAX stock index between 1991 - 1998 (taken from the EuStockMarkets data set from R)

```
EuStockMarkets |>
    as_tsibble() |>
    filter(key == "DAX") |>
    autoplot(value) +
    labs(x = "Business day", y = "Closing price",
     title = "Daily closing prices for DAX stock index")
```

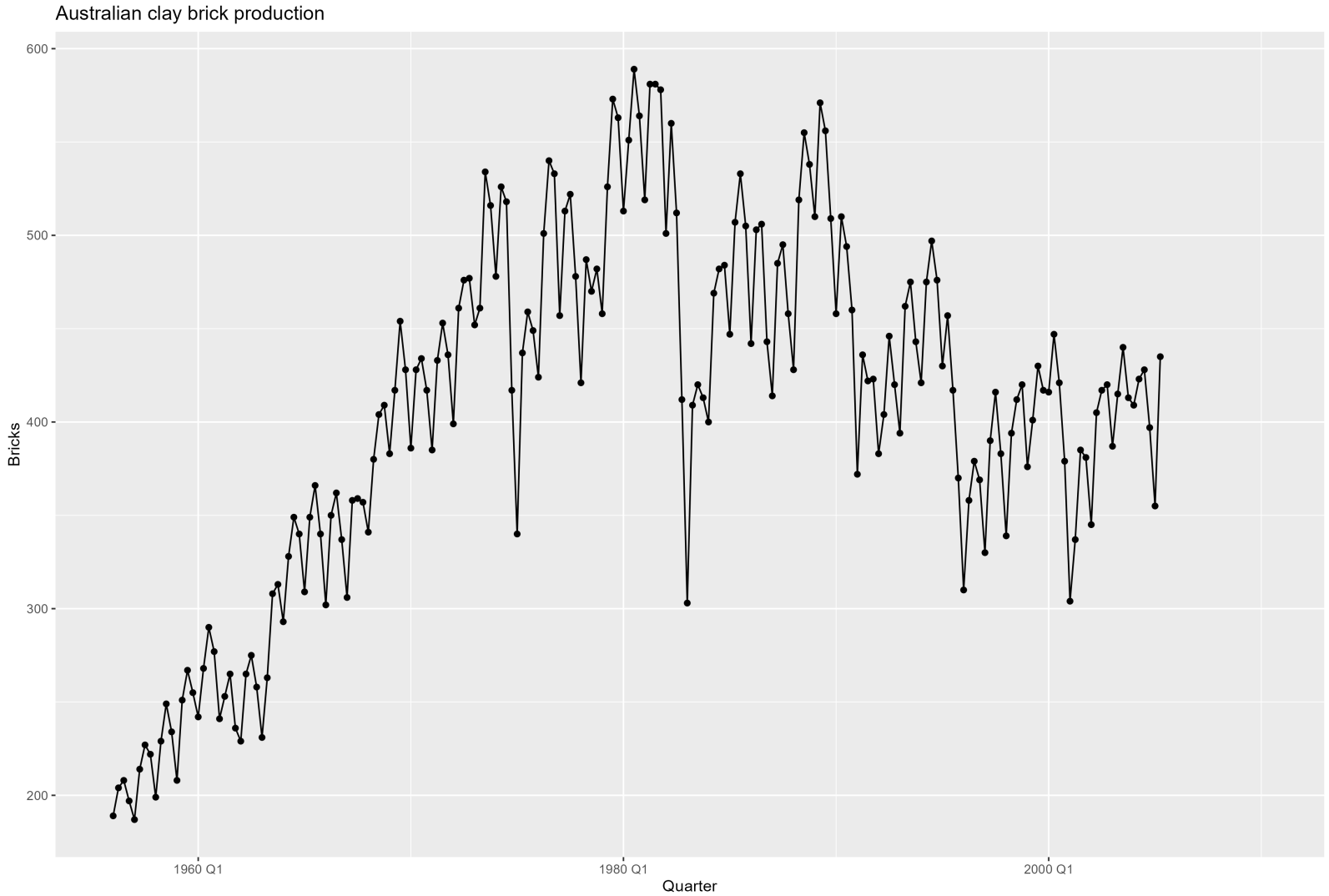



- We have an increasing trend and no seasonality or cycle.

# Trend, season, cycle examples

- Data about quarterly clay brick production between 1956 - 1994 (taken from the *fma* package in R)

```
aus_bricks <- aus_production |>
  select(Bricks)

aus_bricks |>
  autoplot() +
  geom_point() +
  labs(x= "Quarter", y = "Bricks", title = "Australian clay brick production")
```

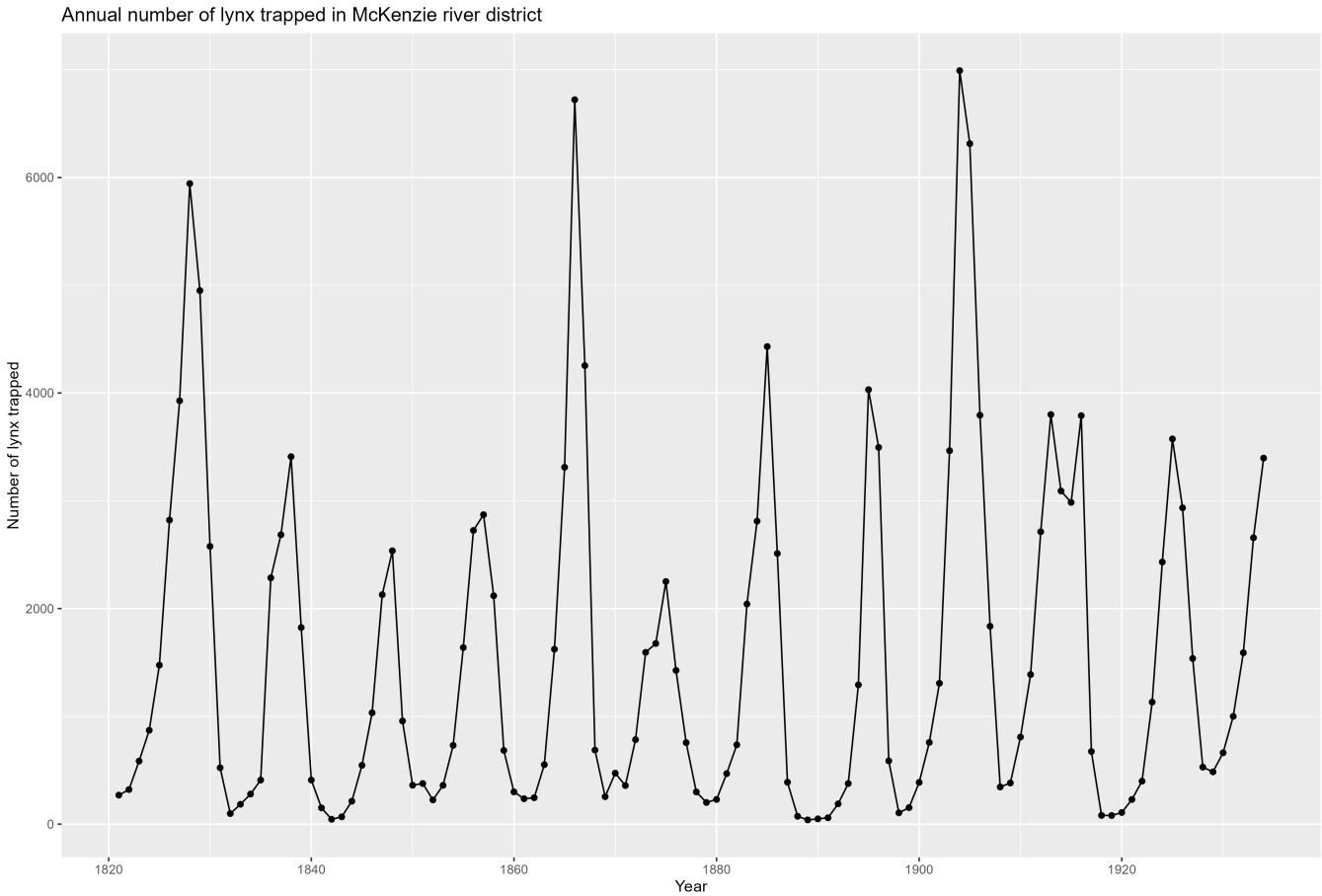



- We have a seasonality with a frequency of a year, but also cycles, showing periods of recession (1975, 1983, 1991) between which the series increases and decreases.

# Trend, season, cycle examples

- Data about annual number of lynx trapped in McKenzie river district of northwest Canada between 1821-1934

```
lynx |>
  autoplot() +
  geom_point() +
  labs(x = "Year", y = "Number of lynx trapped", title =  "Annual number of lynx
      trapped in McKenzie river district")
```

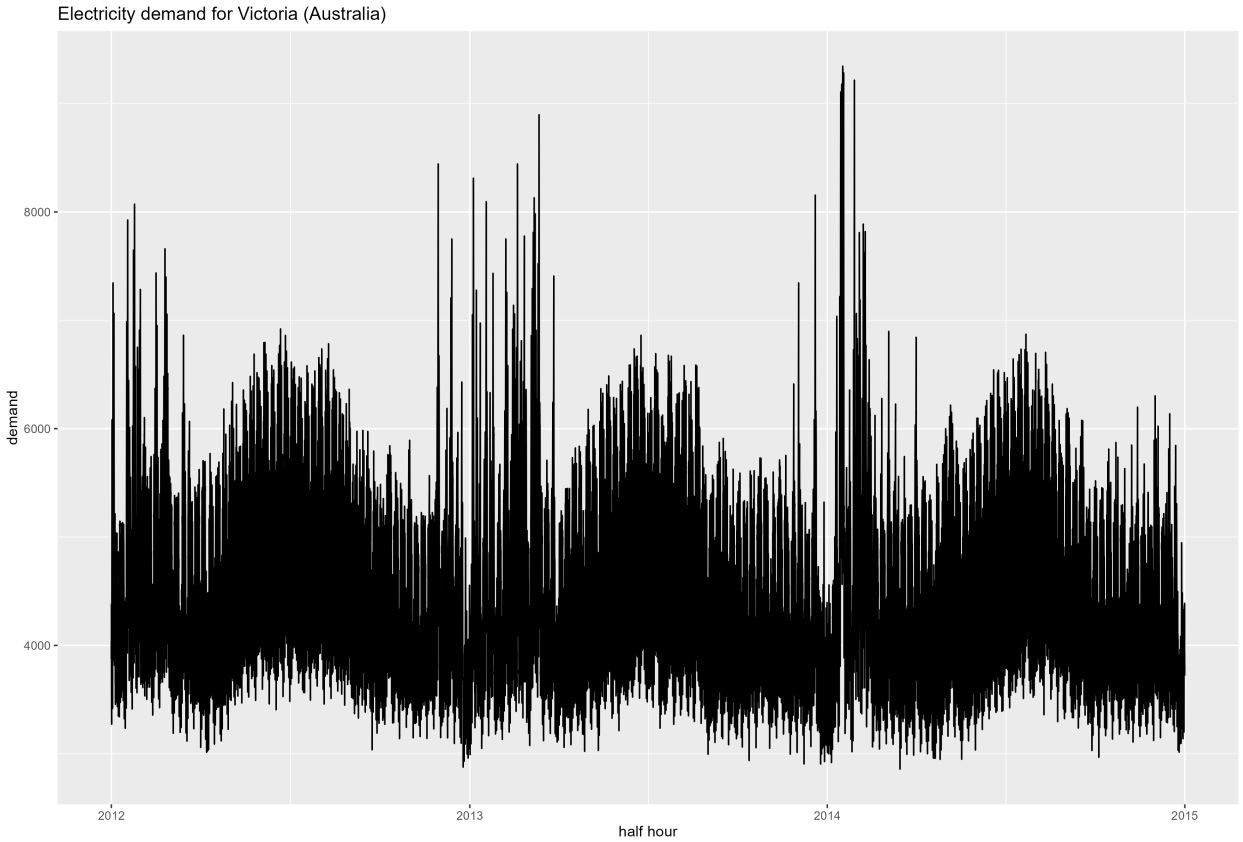


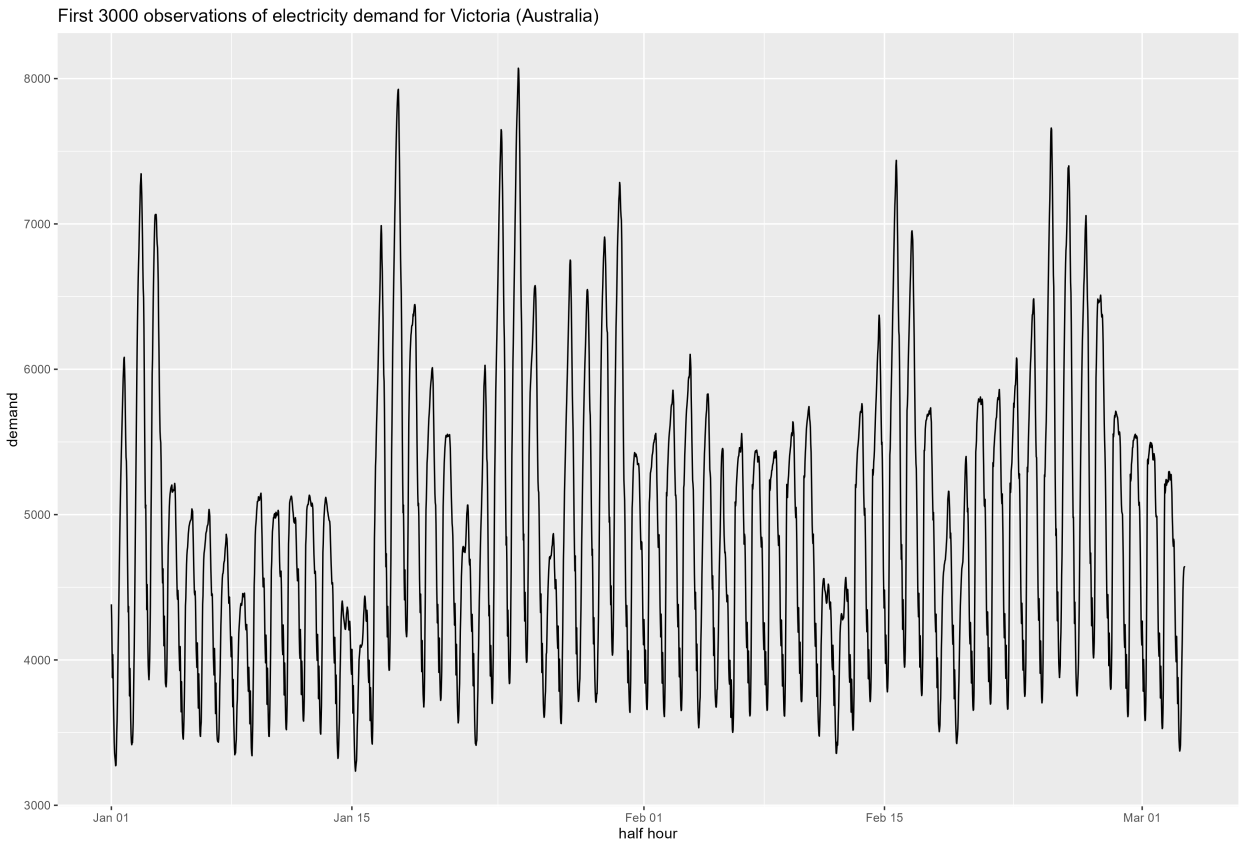

- It might look like seasonal data, but since it is annual, it cannot be seasonal. The length of a "pattern" is several years, so we actually have cycles of 8-11 years.

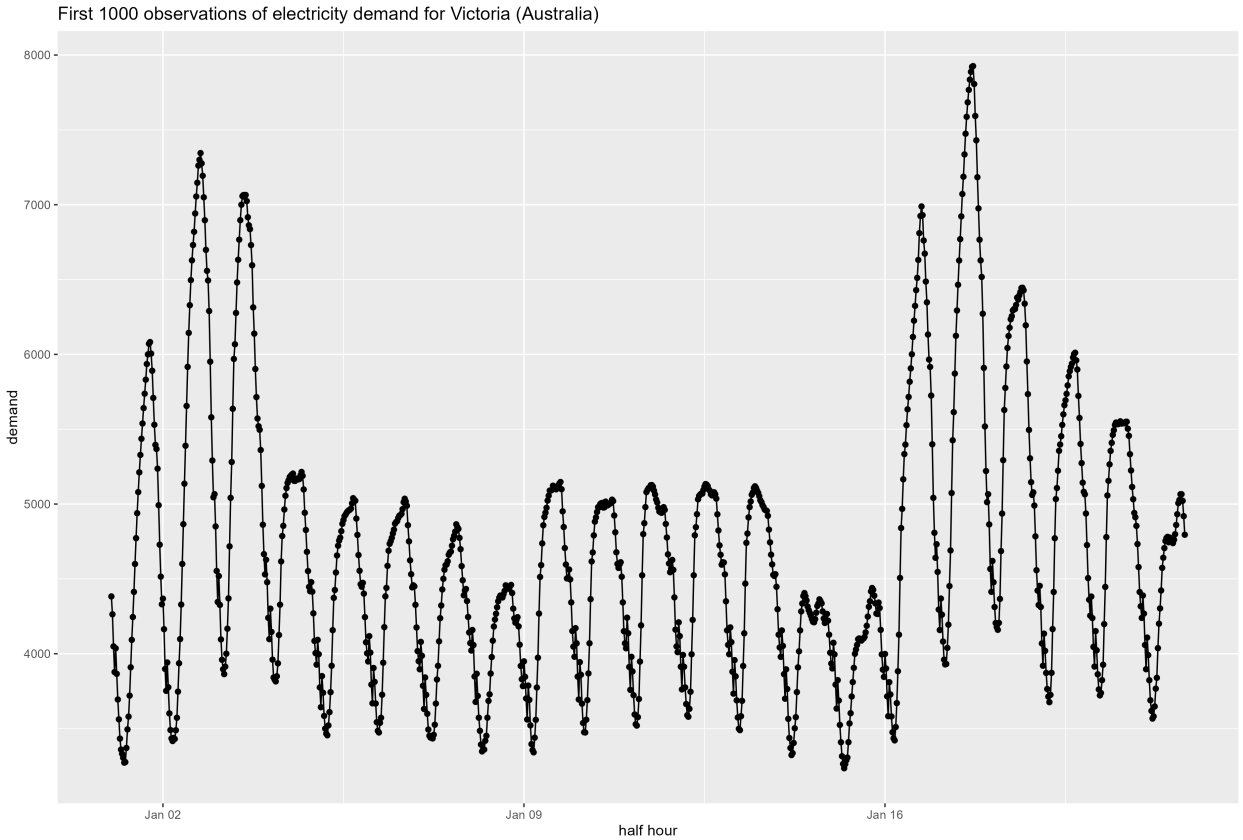
# Trend, season, cycle examples

- Half-hourly electricity demand for Victoria, Australia, between 2012-01-01 and 2014-12-31 (52608 observations). Contains information about the electricity demand and the temperature of Melbourne (taken from package *tsibbledata* in R).

```
vic_elec |>
  autoplot(Demand) +
  labs(x = "half hour", y = "demand", title = "Electricity demand for Victoria (
      Australia)")


vic_elec |>
  head (n = 3000) |>
  autoplot(Demand) +
  labs(x = "half hour", y = "demand", title = "First 3000 observations of
      electricity demand for Victoria (Australia)")

vic_elec |>
  head (n = 1000) |>
  autoplot(Demand) +
  labs(x = "half hour", y = "demand", title = "First 1000 observations of
      electricity demand for Victoria (Australia)")
```


Electricity demand for Victoria (Australia)
8000
6000
4000
demand
2012
2013
2014
2015
half hour


First 3000 observations of electricity demand for Victoria (Australia)
demand
8000
7000
6000
5000
4000
3000
Jan 01
Jan 15
Feb 01
Feb 15
Mar 01
half hour




- There is no trend, but it looks like we have yearly seasons. If we take a closer look at the data (for example, the first 3000 or 1000 observations) we can see that we have a daily season as well. We can have data with two seasons with different length.

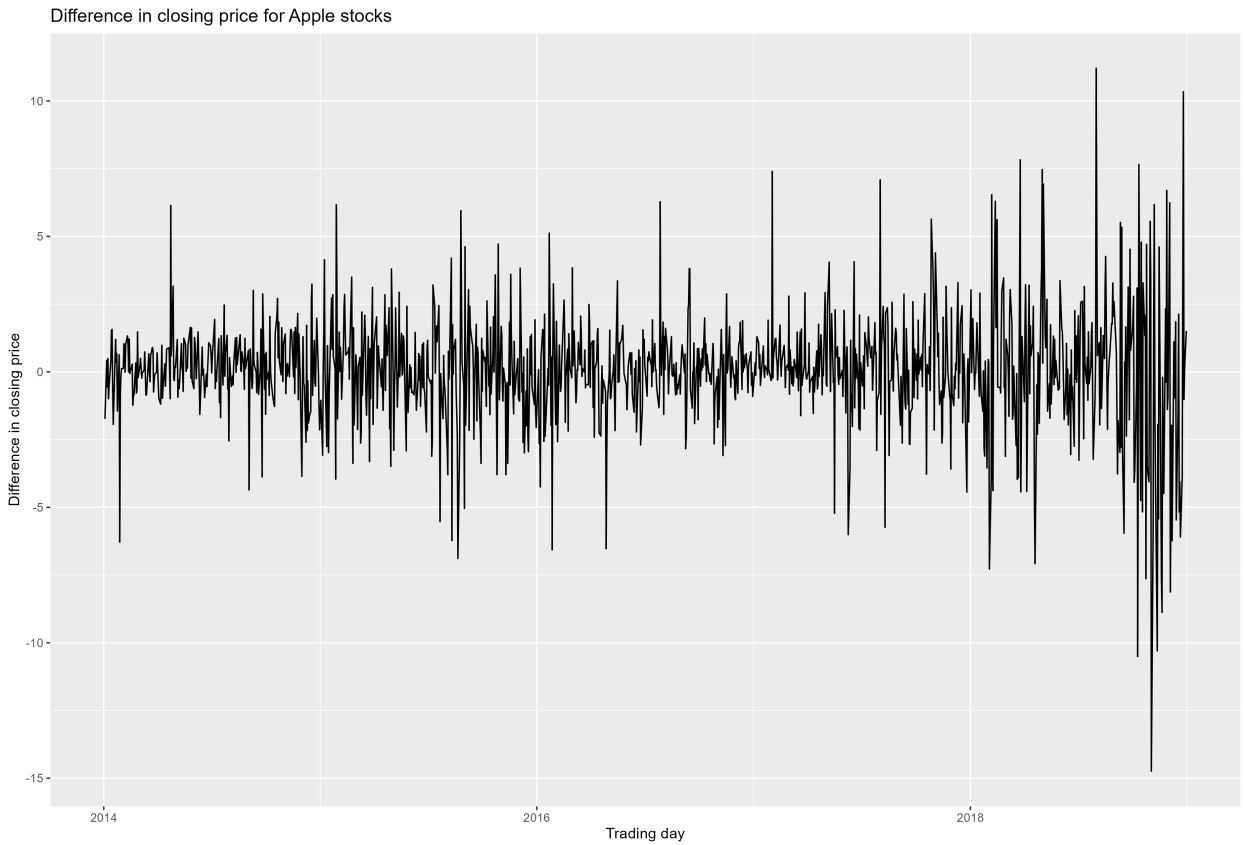
# Trans, season, cycle examples

- When working with trading data, sometimes the actual stock price is less important, than the change from the previous day.

- Let us look at the change for Apple stock prices in the 2014-2018 period.

```
gafa_stock |>
  filter(Symbol == "AAPL") |>
  autoplot(difference(Close)) +
  labs(x = "Trading day", y = "Difference in closing price", title = "Difference
      in closing price for Apple stocks")
```

- **Obs:** the *difference* function simply computes the difference between two consecutive observations (we will talk more about it in a different lecture)

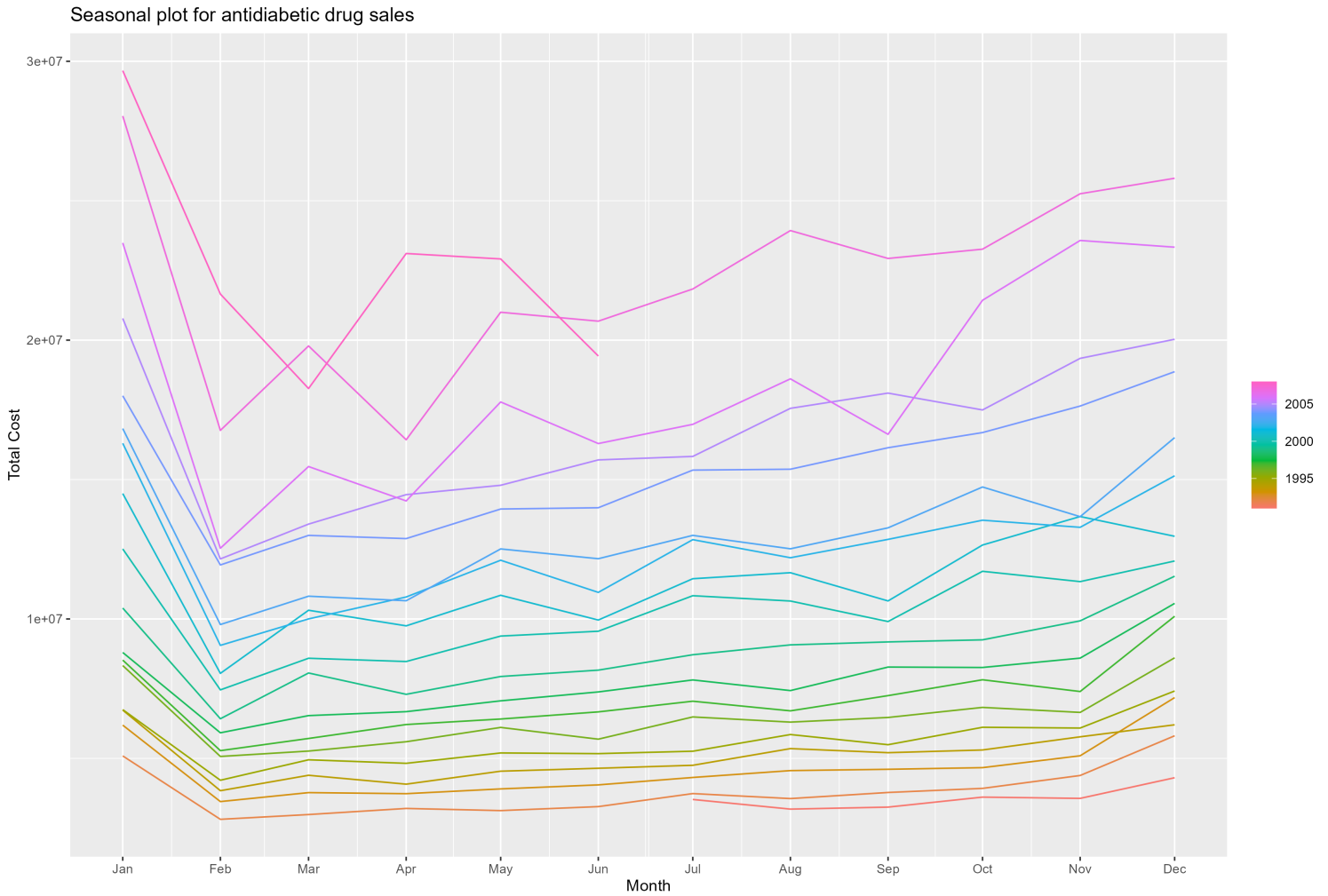



- There is no trend, no cycle, no season. It is stationary.

# Seasonal plots

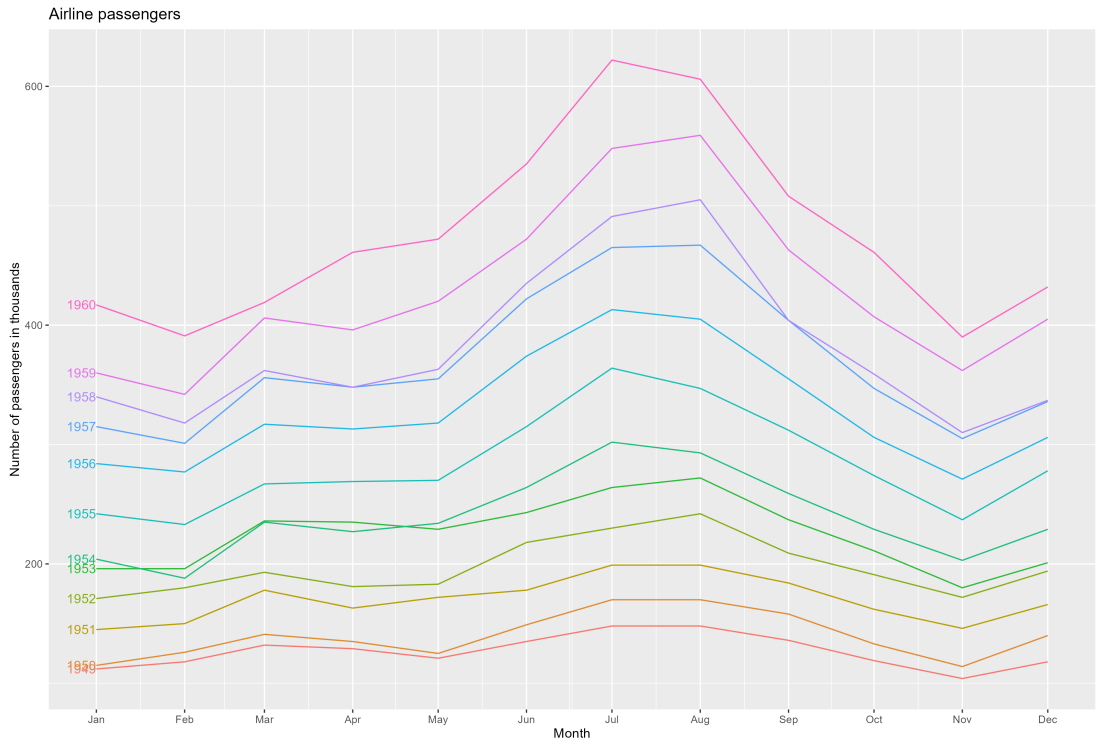
- Once we have identified that there is a seasonality in our time series, we can use a *seasonal* plot to visualize the data.
- A *seasonal* plot simply plots every season *"in parallel"*, so we can see the underlying patterns more clearly and it is easier to detect anomalies or changes in the pattern.
- We can use the *gg_season* for creating seasonal plots.

```
A10 |>
    gg_season(TotalCost) +
     labs(x = "Month", y = "Total Cost", title = "Seasonal plot for antidiabetic
      drug sales")
```

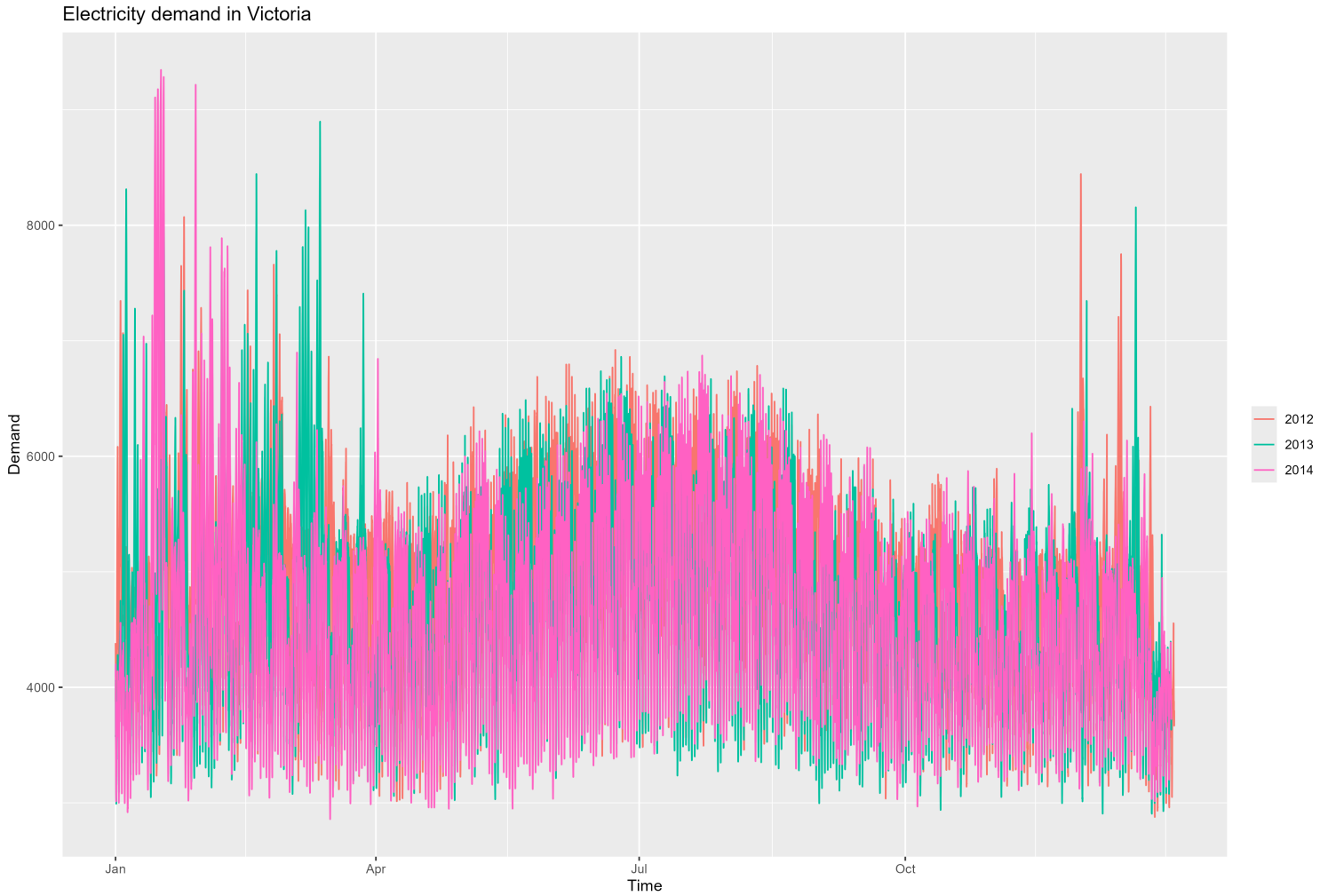

Seasonal plot for antidiabetic drug sales
Total Cost
3e+07
2e+07
1e+07
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
Month
2005
2000
1995

- We can see that the lines are mostly parallel $\rightarrow$ there is a seasonal pattern in data.
- We can see that the values in January are high, which drop for February.
- We can see that in 2008 March there was a small number of sales (in general March has higher sales than February).
- Also we can see that over the years the sales have increased in general (more recent lines are above the older ones - this is actually the increasing trend of the time series).

```
AirPassengers |>
  as_tsibble() |>
  gg_season(value, labels = "left") +
  labs (x = "Month", y = "Number of passengers in thousands", title = "Airline
      passengers")
```

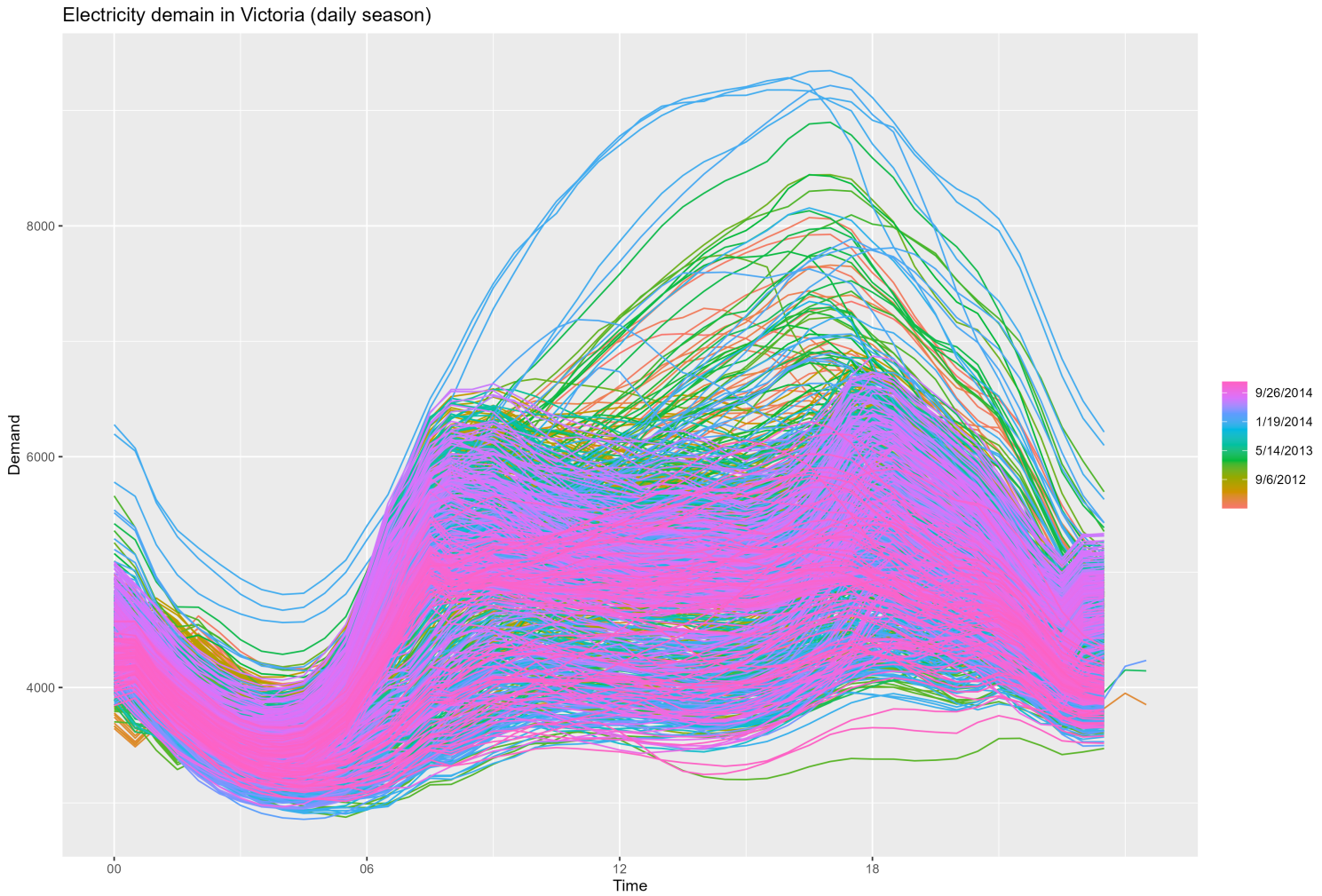


```
vic_elec |> gg_season(Demand)
```



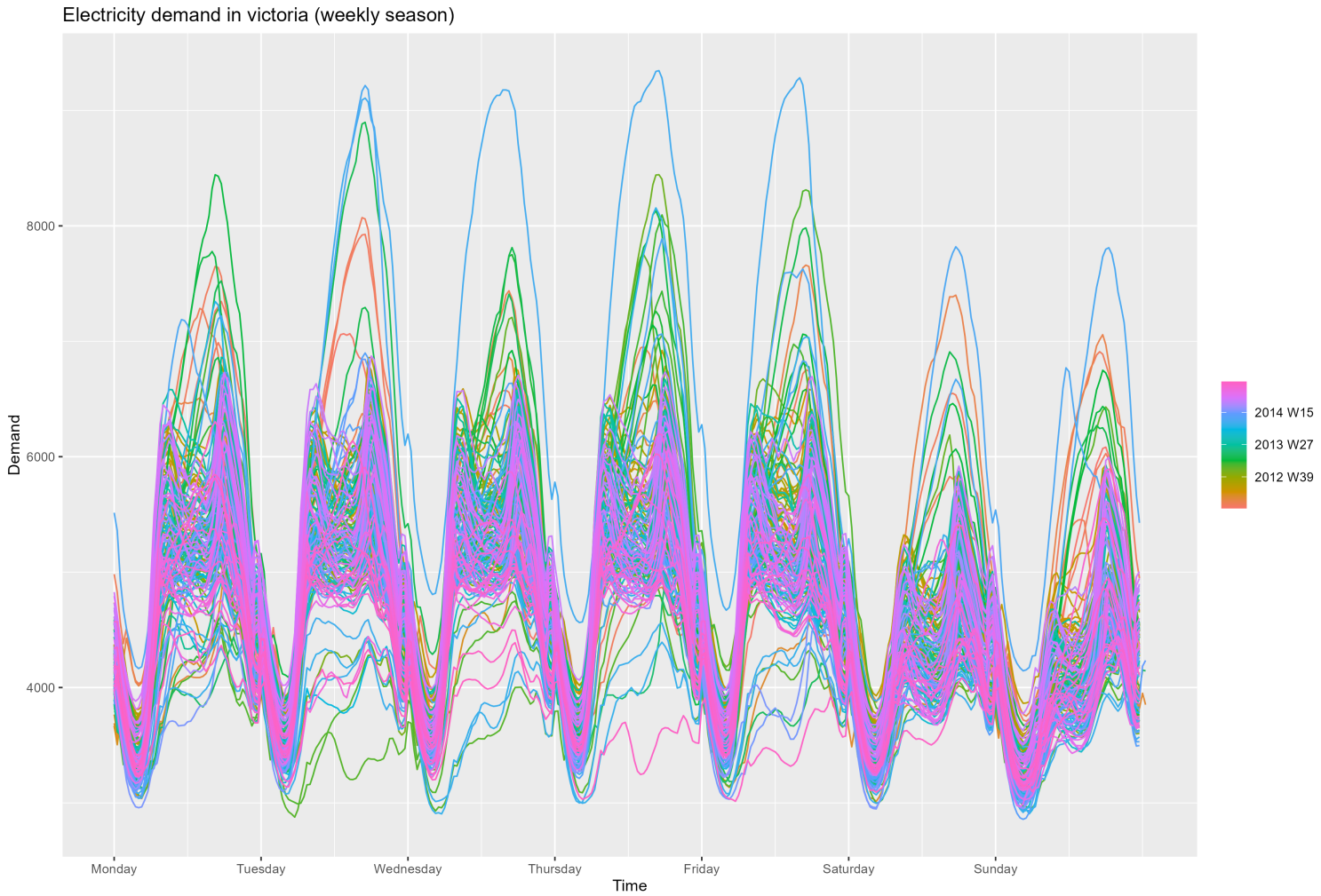
- We saw from the time series plot that the *vic_elec* data has two seasons. We can plot the daily season as well.

```
vic_elec |>
    gg_season(Demand, period = "day") +
    labs(title = "Seasonal plot for daily seasons of electricity demand")
```


Electricity demain in Victoria (daily season)
Demand
8000
6000
4000
00
06
12
18
Time
9/26/2014
1/19/2014
5/14/2013
9/6/2012

- If data has a daily seasonality, we might want to check if it has weekly seasonality as well:

```
vic_elec |>
  gg_season(Demand, period="week") +
  labs(title = "Electricity demand in victoria (weekly season)")
```


Electricity demand in victoria (weekly season)
Demand
8000
6000
4000
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
Time
2014 W15
2013 W27
2012 W39

- Let's see what happens if we plot a time series without seasons, for example the *DAX stock prices* time series.

```
EuStockMarkets |>
  as_tsibble() |>
  filter(key == "DAX") |>
  gg_season()
```

- We will get an error due to missing values.

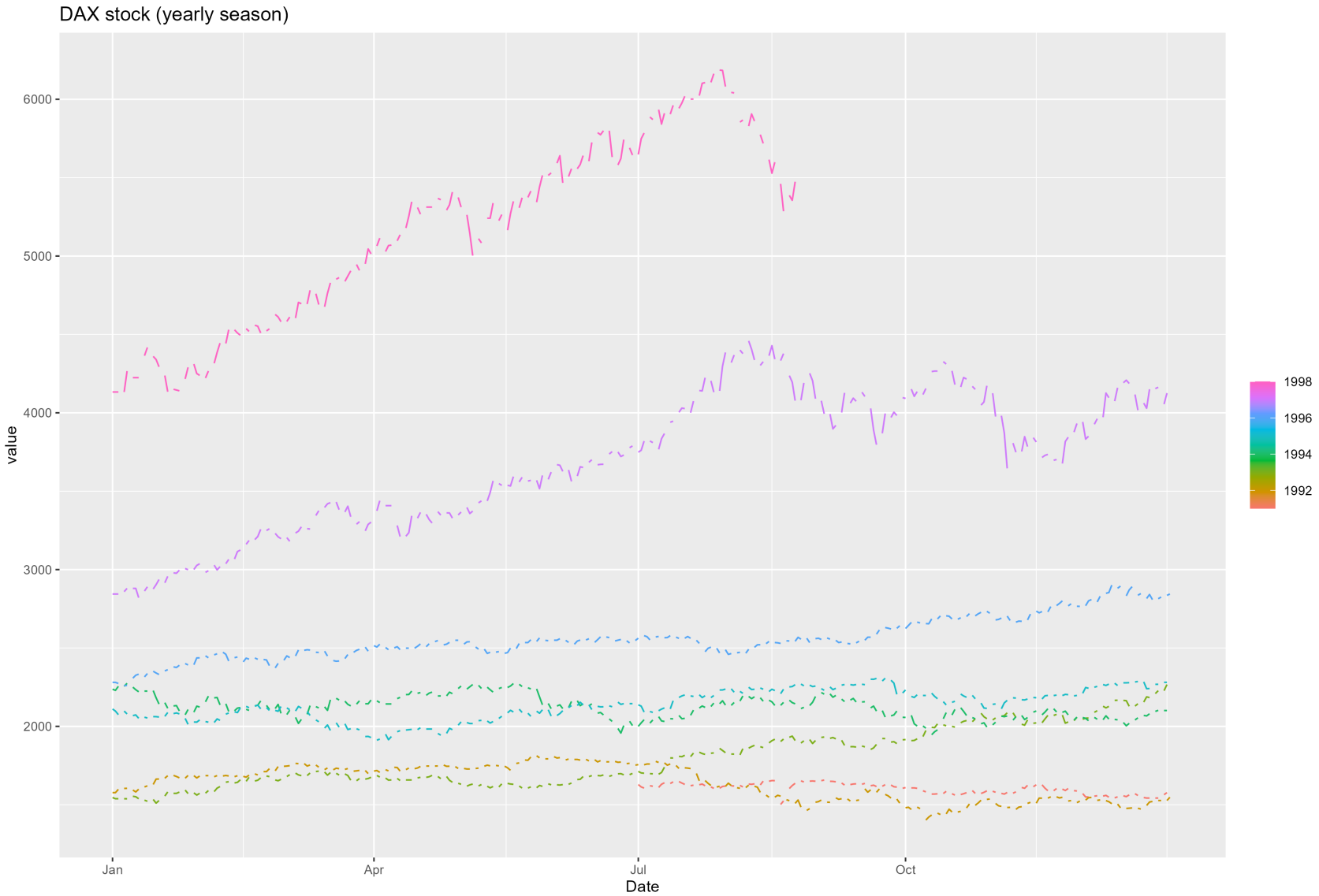
# Missing data

- In case a time series, we can have two types of missing data: implicit missing data (when actual time stamps are missing) or explicit missing data (when time stamps are present, but there are no values associated to them)
- If we have implicit missing values in the time series *gg_season* will not work.
- If we have explicit missing values in the time series, *gg_season* will just leave an empty space in the plot.
- We can transform implicit missing values into explicit ones using the *fill_gaps* function.

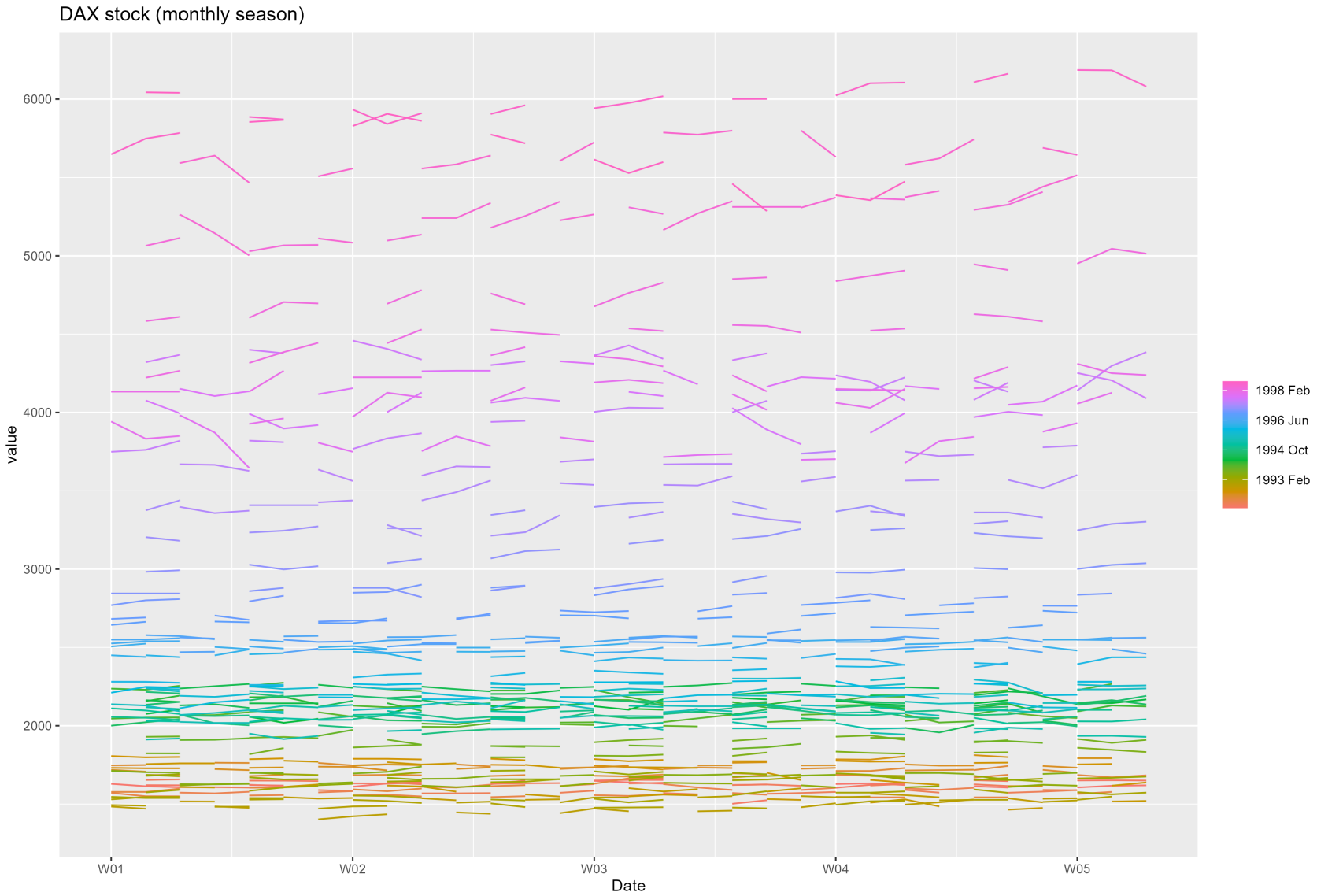
# Transforming missing values

- *Missing data* can be found frequently at day-level time series, where only business days are considered (for example stock market information).

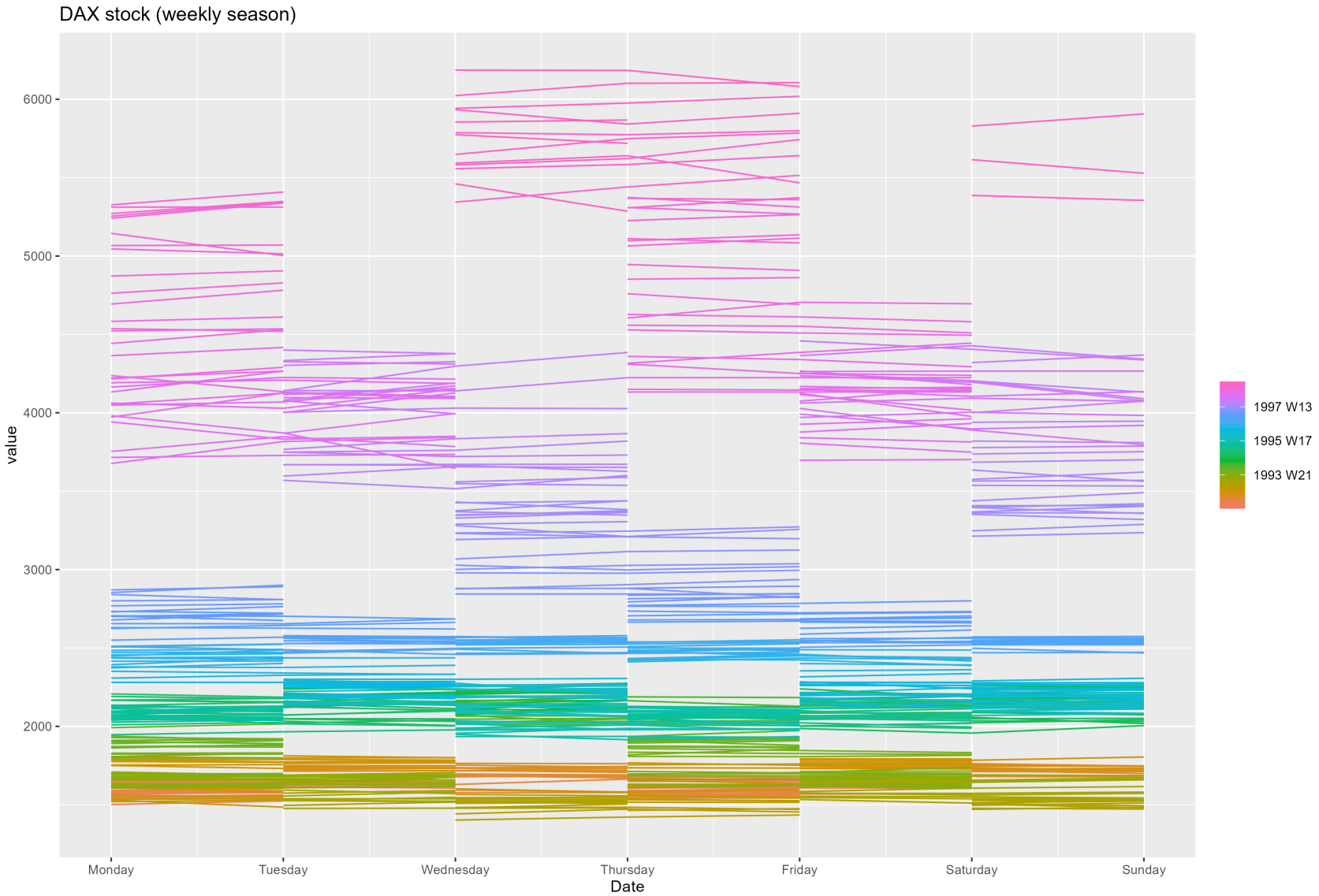
```
EuStockMarkets

EuStockMarkets |>
  as_tsibble() |>
  filter(key == "DAX") |>
  mutate(Date = as.Date(index)) |>
  update_tsibble(index = Date) |>
  select(Date, value) -> DaxStock

DaxStock
DaxStock |> fill_gaps(.full = TRUE) -> DaxStockNoMissing
```



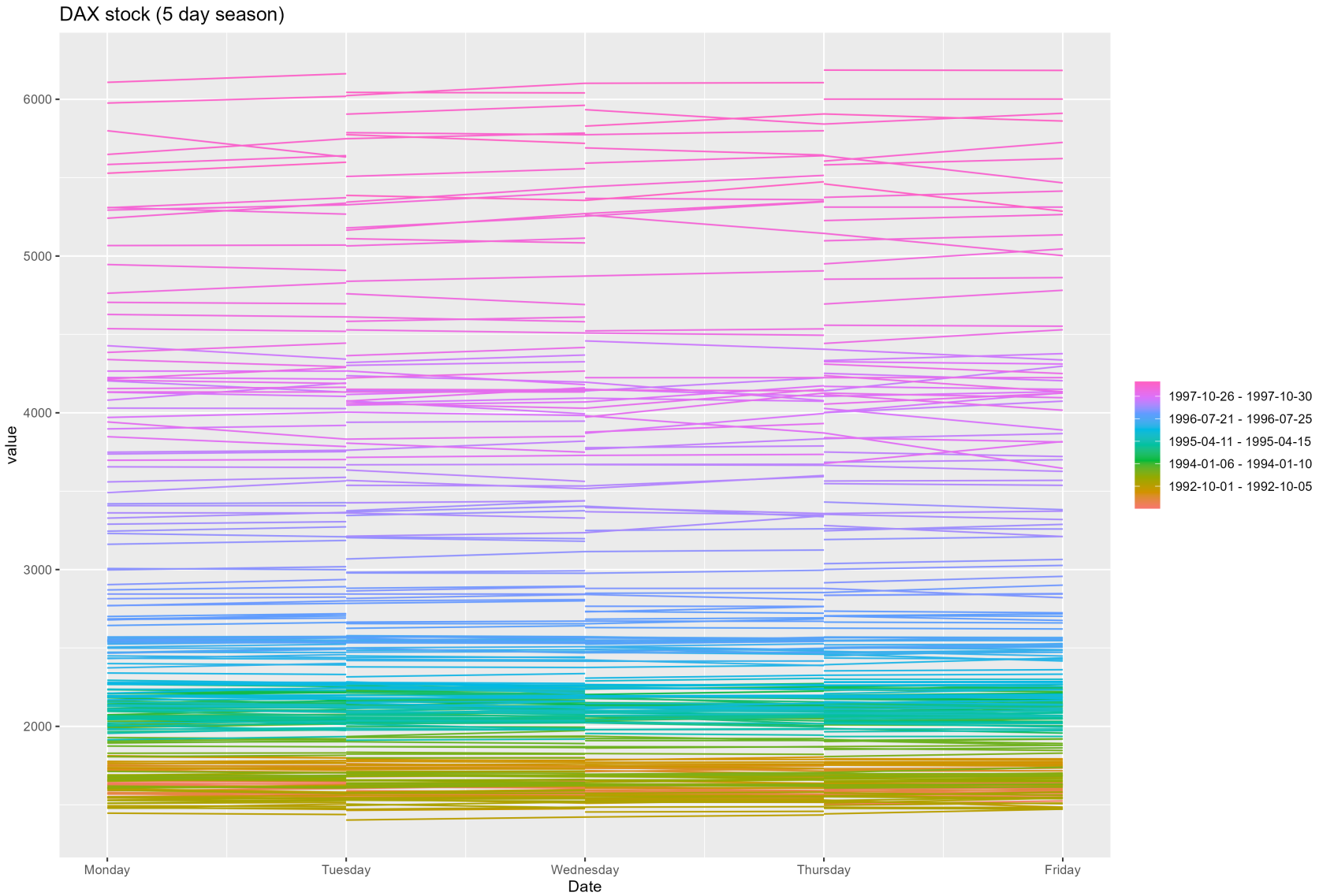

- The default yearly seasonality




- Monthly seasonality




- Daily seasonality

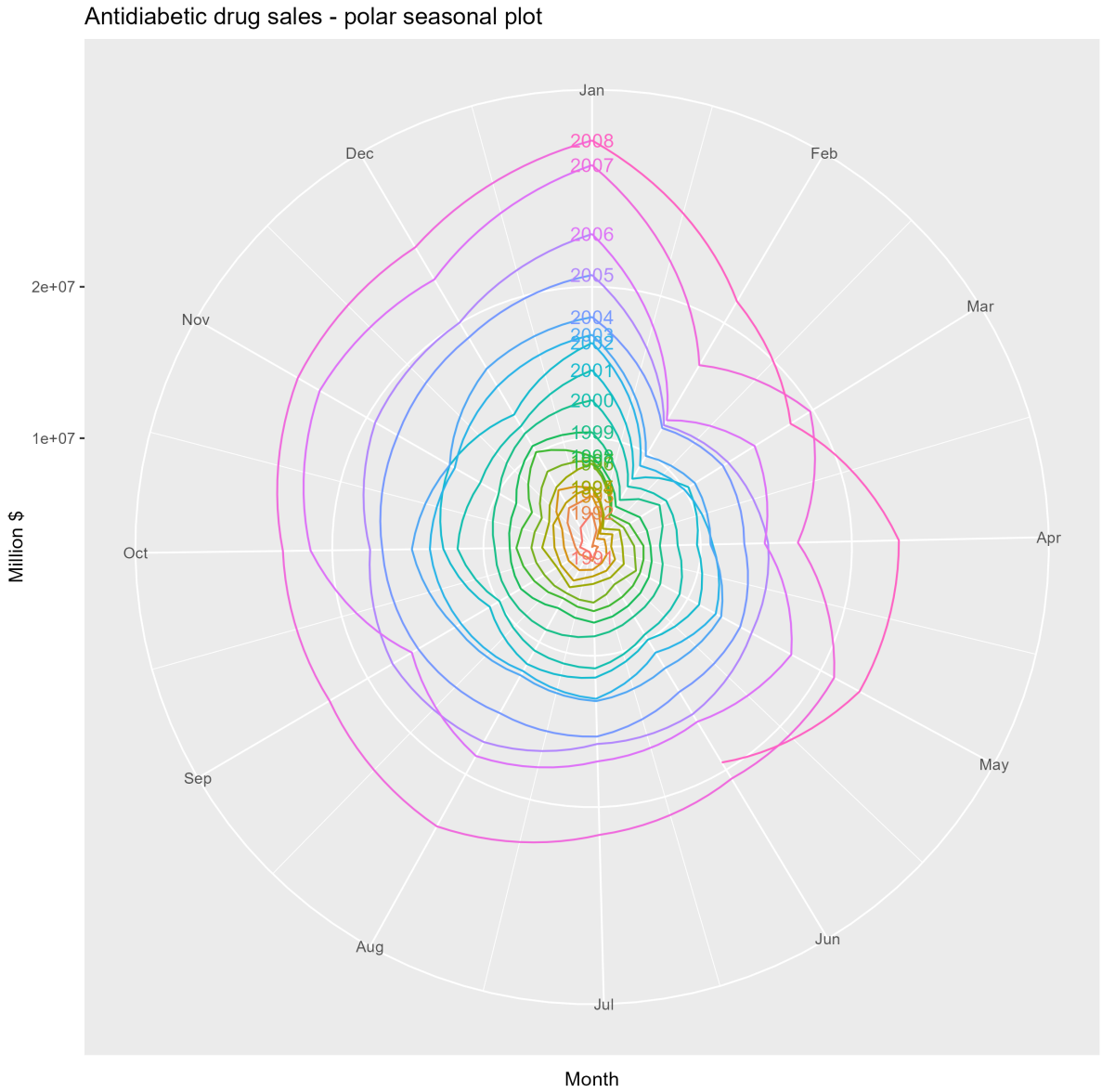



- 5 day long seasonality

- Regardless of the seasonal period, the DAX Stock data does not have seasons.

- Another option when creating seasonal plots is to use polar coordinates for it, by setting the value of the *polar* parameter to TRUE.

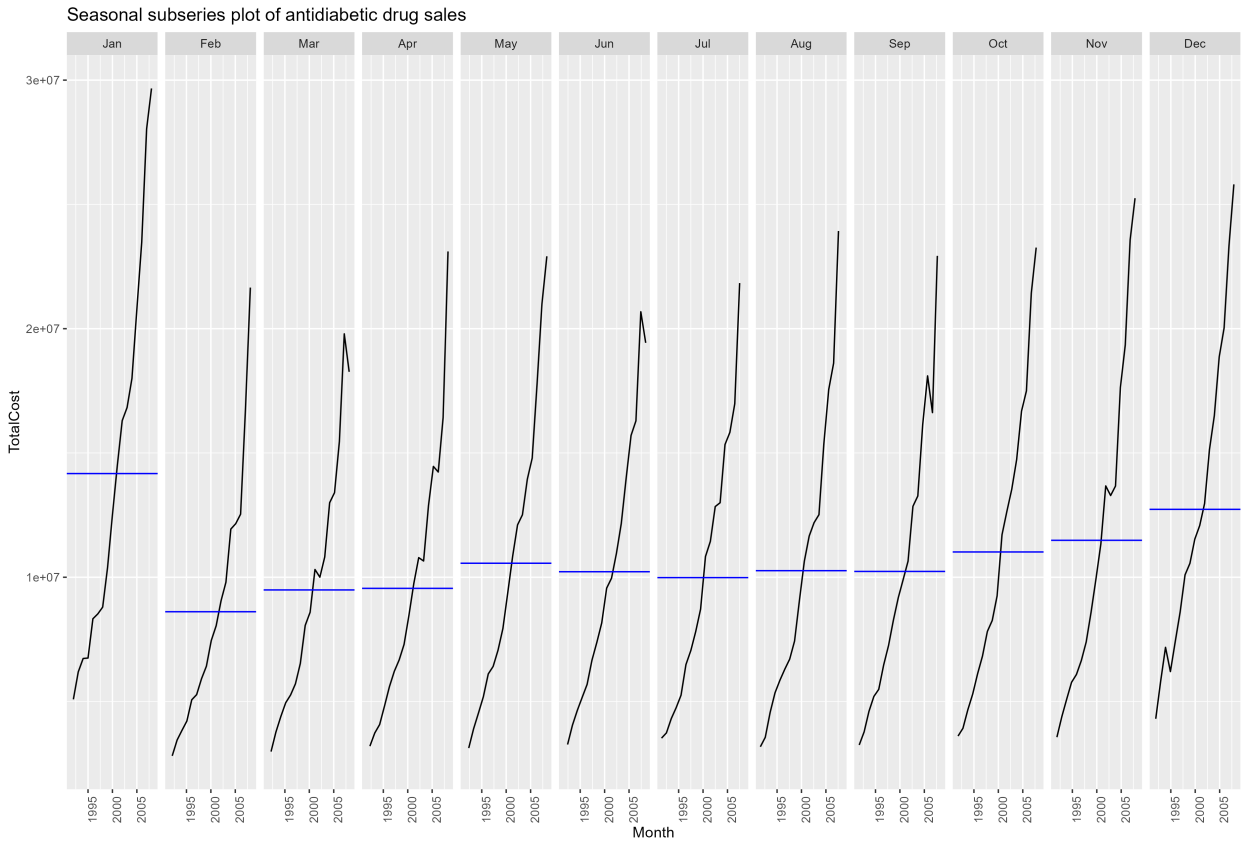
```
gg_season(A10, labels="left", polar = TRUE) +
 ylab("Million $") +
 ggtitle("Antidiabetic drug sales - polar seasonal plot")
```


Antidiabetic drug sales - polar seasonal plot
Jan
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Oct
Nov
Dec
2e+07
1e+07
Million $
2008
2007
2006
2005
2004
2001
2000
1999
Month

# Seasonal subseries plot

- Another plot suitable for seasonal data is the *seasonal subseries plot*, which creates a separate plot for each season of your data for points from successive time periods. For each subplot, a horizontal line is also drawn which is the mean of the values.
- Such a plot can reveal:
  - Changes within a given season over time
  - Changes between different seasons
- On the other hand, it is complicated to compare the time series to each other using the seasonal subseries plot.
- You can create a seasonal subseries plot with the *gg_subseries* function.

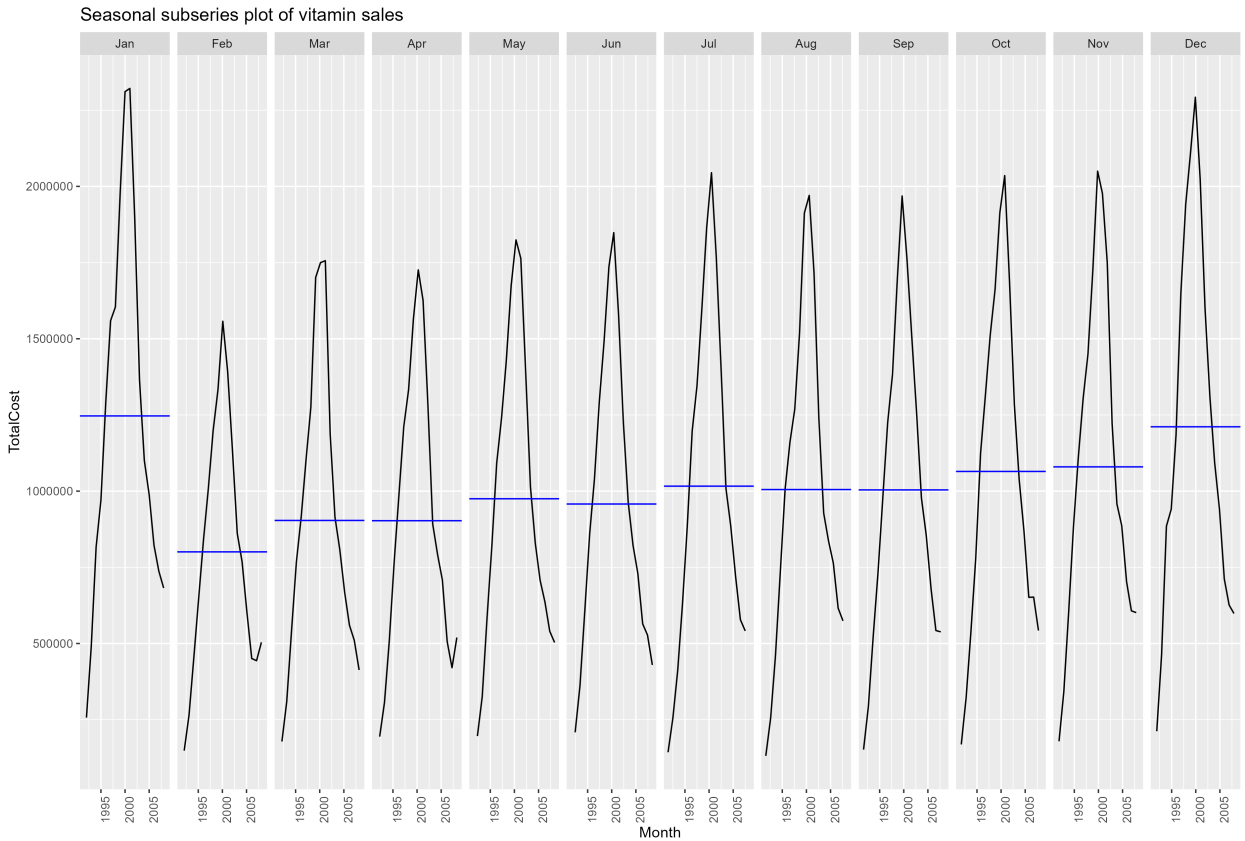
```
a10 |>
    gg_subseries() +
    labs(title = "Seasonal subseries plot of antidiabetic drug sales")
```



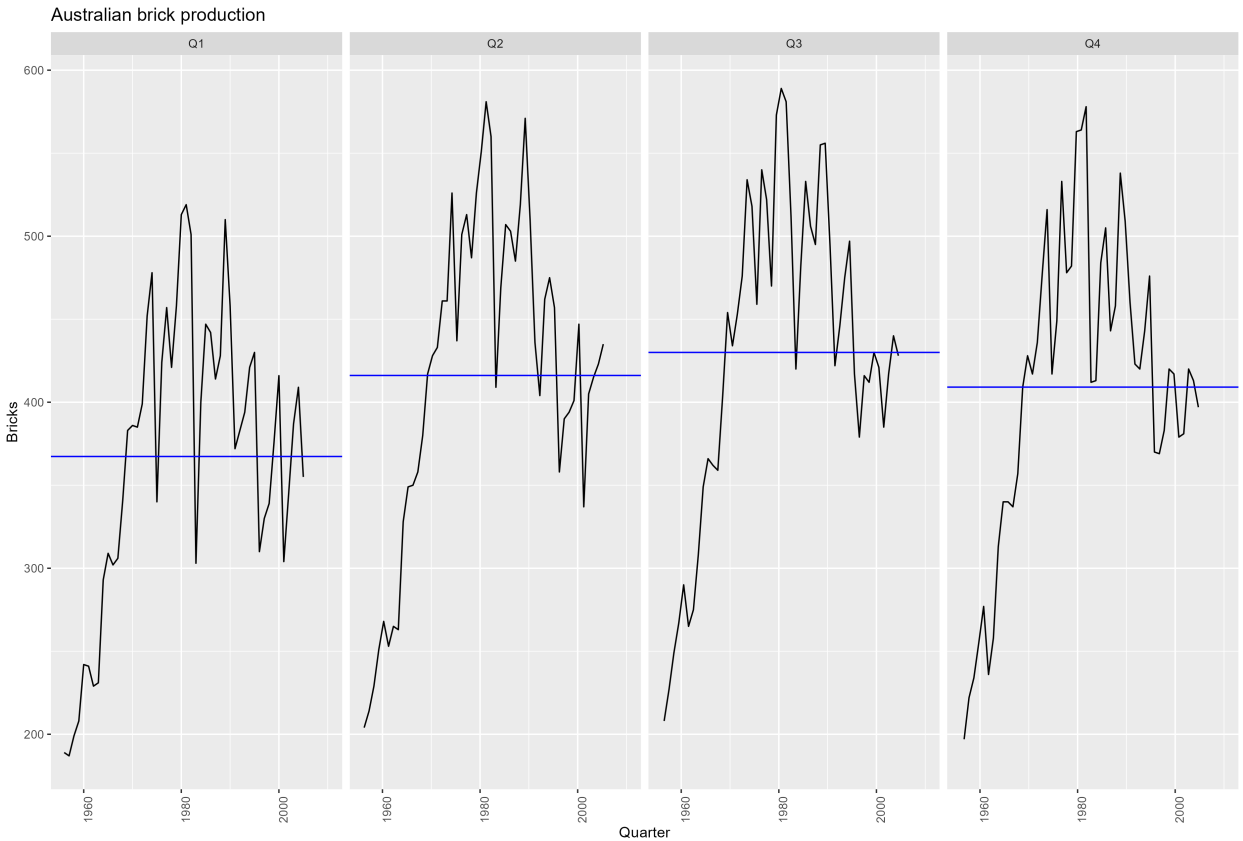
- We can see that there are months for which the trend is constantly increasing (ex. January), but for March and June it was decreasing in the last years.

- Also, there are months for which in past there we short-term drops (ex: September).

```
a11 |>
    gg_subseries() +
    labs(title = "Seasonal subseries plot of vitamin sales")
```



```
aus_bricks |>
    gg_subseries() +
    labs(title = "Australian brick production")
```



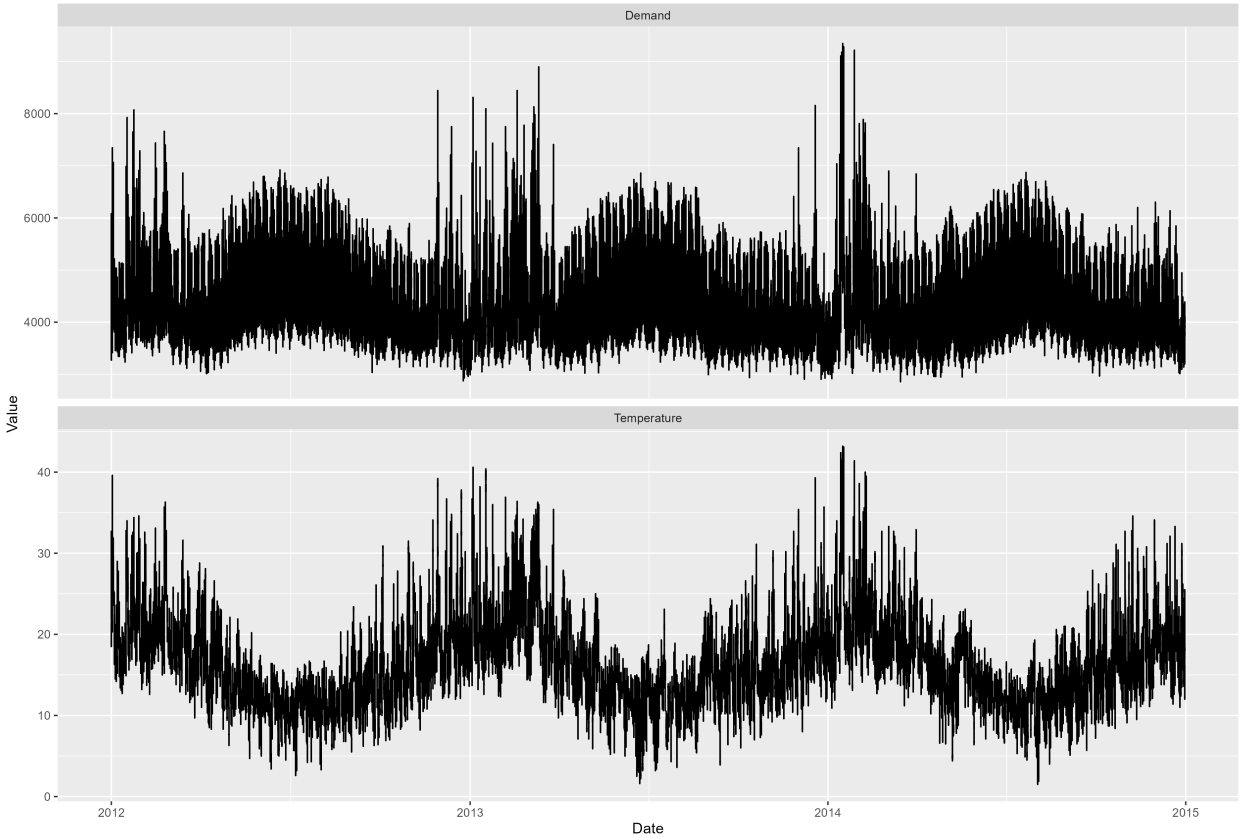
# Scatter plot

- Time series, seasonal and seasonal subseries plots can be used to visualize one single time series. Sometimes we want to see relations between time series.
- For example: the *vic_elec* time series contains three attributes:
  - Demand - Total electricity demand for Victoria, Australia (every half-hour)
  - Holiday - TRUE if the current day is a Holiday, FALSE otherwise
  - Temperature - temperature for Melbourne (every half hour)
- Demand and temperature are time series, so we can plot them.

- *autoplot* cannot handle two time series, unless they are in the long data format (and *vic_elec* is in the wide format). So we will first transform it, using *pivot_longer*.

- Now we could autoplot it, but since the scales are so different (Temperature is between 1.5 and 43.20, while Demand is between 2858 and 9345) Temperature will not really be visible.
- We should plot them in two different facets and for this we need *ggplot*.

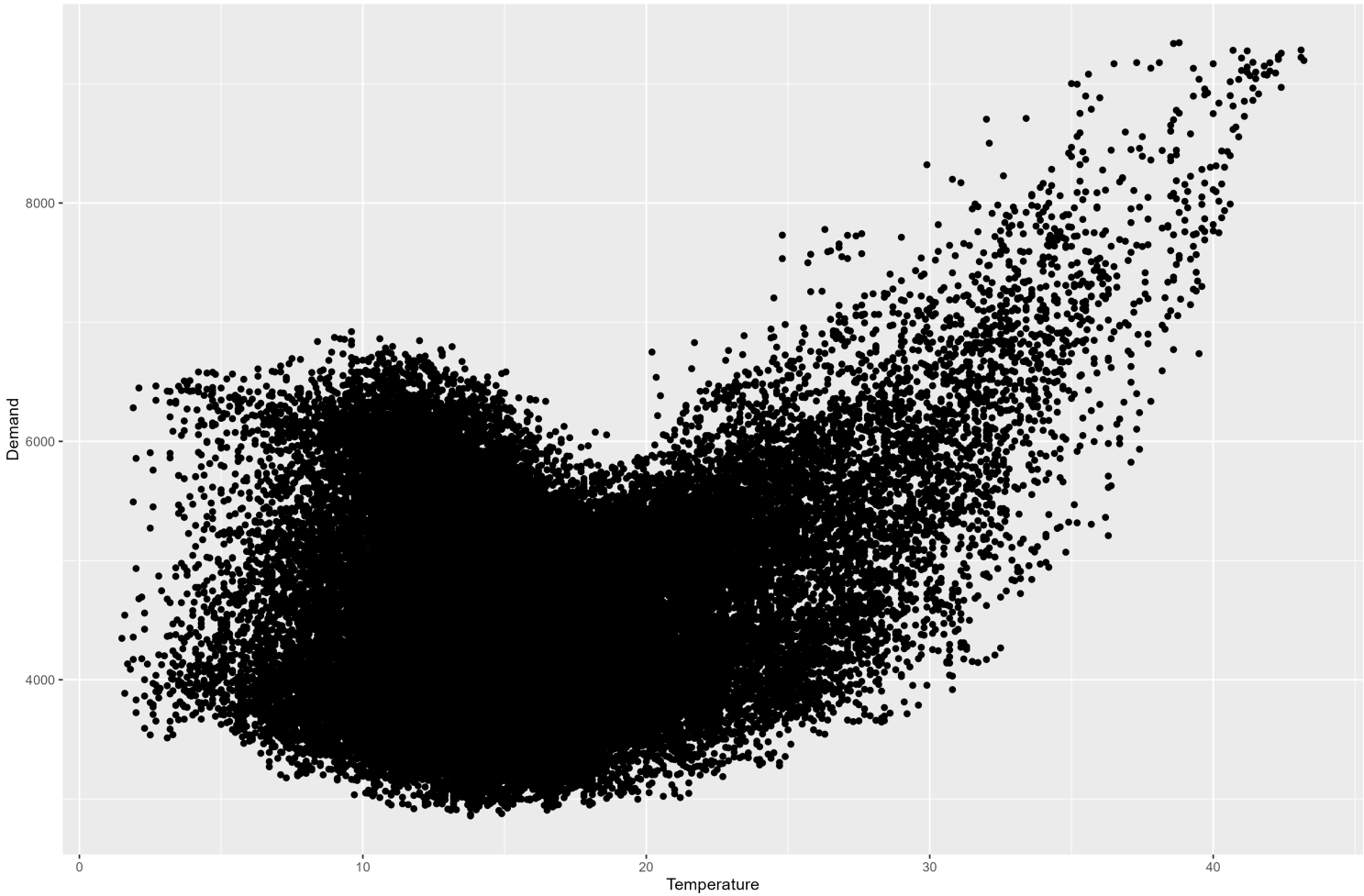
```
pivot_longer(vic_elec, cols = c("Demand", "Temperature"),
    names_to = "Attribute", values_to = "Value") -> vic_elec_long
autoplot(vic_elec_long)
ggplot(data = vic_elec_long) +
  geom_line(mapping = aes(x = Date, y = Value)) +
  facet_wrap(vars(Attribute), scales= "free_y", nrow = 2, ncol = 1)
```


Demand
Temperature
Value
8000
6000
4000
40
30
20
10
0
2012
2013
2014
2015
Date

- While we might be able to see that there are some common patterns in both series, it is more helpful if we plot the two series together, one against the other in a scatter plot, where each point of the plot is a (Temparature, Demand) pair of observations.

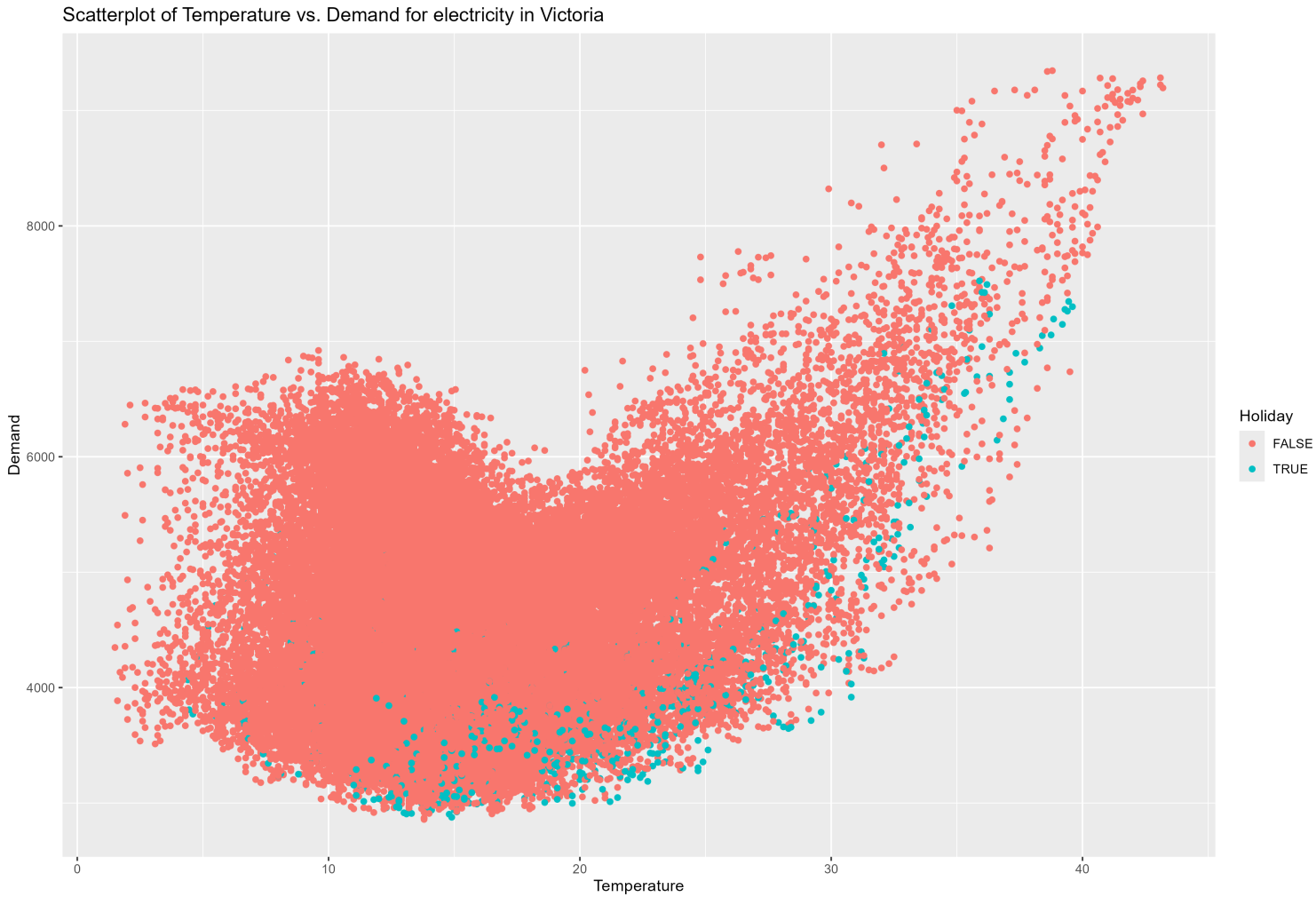
```
ggplot(data = vic_elec) +
   geom_point(mapping = aes(x = Temperature, y = Demand)) +
    labs(title = "Scatterplot of Temperature vs. Demand for electricity in
      Victoria"
```

Scatterplot of Temperature vs. Demand for electricity in Victoria



- Remember, *vic_elec* contained an attribute denoting if the current day is a holiday or not. By adding a color aesthetic for this, we can see whether the relation between Temperature and Demand is the same on holidays as in regular days.

```
ggplot(data = vic_elec) +
  geom_point(mapping = aes(x = Temperature, y = Demand, color = Holiday))
```


Scatterplot of Temperature vs. Demand for electricity in Victoria
Demand
8000
6000
4000
0
10
20
30
40
Temperature
Holiday
FALSE
TRUE

# Correlation

- Scatterplots help us visualize relationships between time series. In the previous example we could see that electricity demand increases when temperatures increase (people turn on air-conditioning), but to a limited extent, demand increases when temperatures are low (due to some using electricity for heating).
- If we want a numerical value for how strong is the relationship between two variable we can compute the *correlation coefficient*.
- Correlation takes values between -1 (strong negative correlation) and 1 (strong positive correlation). Values around 0 mean that the variables are not correlated.

# Correlations

- Correlations in R can be computed with the *cor* function.
- There are several types of correlations, the *cor* function can compute 3 of them:
  - Pearson (the default) - measures linear relationship
  - Spearman - measures the strength of monotonic association between two variables
  - Kendall - Similar to Spearman (as in works on ranks), but it is better when there are few data points and many rank ties.

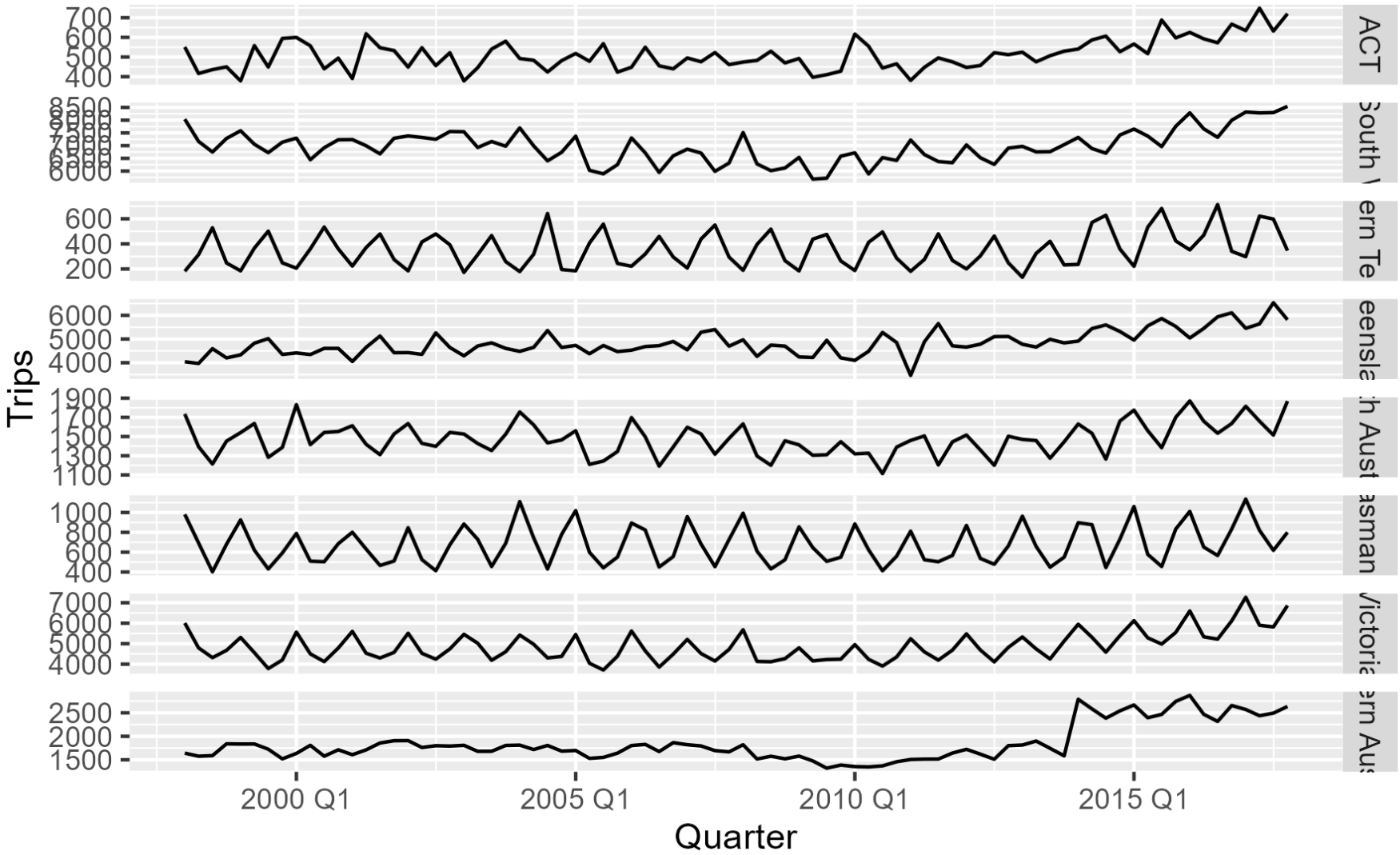
- Pearson correlation for the electricity demand and temperature data is 0.26, which is quite low. The reason for this is that the relationship between these two variables is not linear.
- Spearman and Kendall correlations are even lower (0.11 and 0.08).
- This shows that it is important to plot the data as well, not just rely on numerical correlation values.

- Correlations should be used very carefully when we work with time series!!!
- Two time series (for which we compute correlations) are actually connected by time, which might influence (increase) the results.
- This is especially true in case of time series which have a trend.
- Spurious Correlations: `https://www.tylervigen.com/spurious-correlations`
- Nice reading and example about what is the problem when computing correlation between two trended time series: `https://www.svds.com/avoiding-common-mistakes-with-time-series/`

- Sometimes we have data which consists of several time series (one example is the *vic_elec* data set containing two time series).
- In this case, we might want to create scatter plots for every combination of variables. We did ithis to *vic_elec*, let's see now an example with more time series.
- For example: the *tourism* data set contains 304 time series (total of 24320 observations), each of them representing the quarterly visitor nights for different regions of Australia, between 1998 - 2017. It contains the following attributes:
  - Quarter
  - Region (76 different values)
  - State (8 different values)
  - Purpose (4 possible values: Business, Holiday, Visiting, Other)
  - Number of trips
  - The keys of the data set are the Region, State and Purpose (but Region defines the State automatically)

- To reduce the number of time series, let's group the observations by state (which means that we will have one observation per state) and compute the total number of trips for each of them.
- Now we only have 640 observations (8 time series * 4 quarters * 20 years)
- Let's plot the 8 time series (using facets)

```
visitors <- tourism |>
  group_by(State) |>
  summarize(Trips = sum(Trips))

ggplot(data = visitors) +
  geom_line(mapping = aes(x = Quarter, y = Trips)) +
  facet_grid(vars(State), scales = "free_y")
```

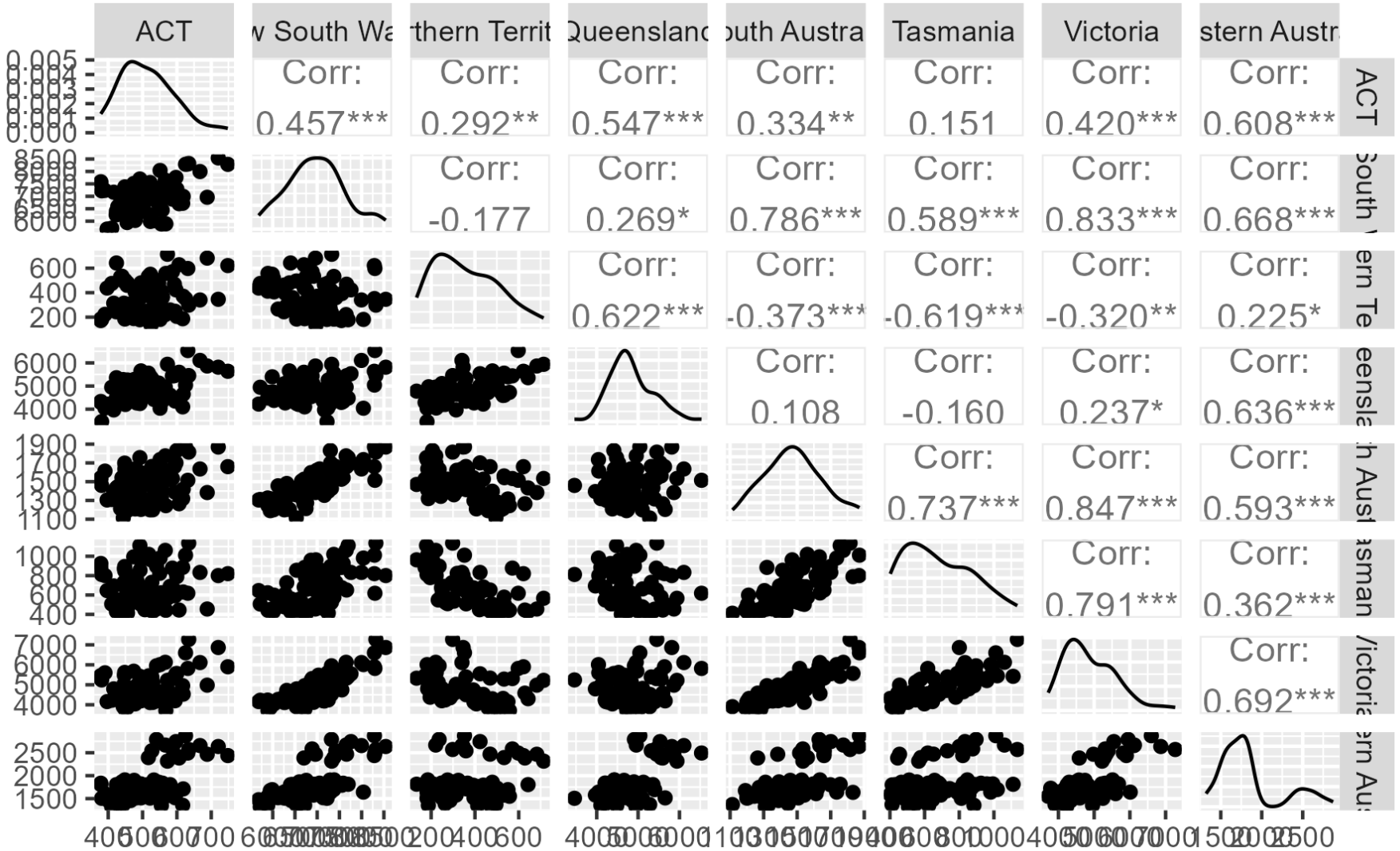

Trips
Quarter
2000 Q1
2005 Q1
2010 Q1
2015 Q1
ACT
Victoria

- We might want to visualize scatter plots between pairs of variables, which can be done with the *ggpairs* function, for which we need to install the *GGally* package.

- *ggpairs* takes the time series to be considered as columns (needs wide format) but we have the long one. So we will use *pivot_wider* to transform it first.

- *ggpairs* by default considers all attribute pairs. But we have the Quarter attribute which should not be considered. So we will use the *columns* parameter to show which attributes to consider.

```
visitors_wide <- pivot_wider(visitors, values_from= Trips, names_from=State)

ggpairs(visitors_wide, columns = 2:9)
```


ACT
w South Wa
thern Territ
Queensland
outh Austra
Tasmania
Victoria
stern Austr
Corr:
0.457***
0.292**
0.547***
0.334**
0.151
0.420***
0.608***
-0.177
0.269*
0.786***
0.589***
0.833***
0.668***
0.622***
-0.373***
-0.619***
-0.320**
0.225*
0.108
-0.160
0.237*
0.636***
0.737***
0.847***
0.593***
0.791***
0.362***
0.692***

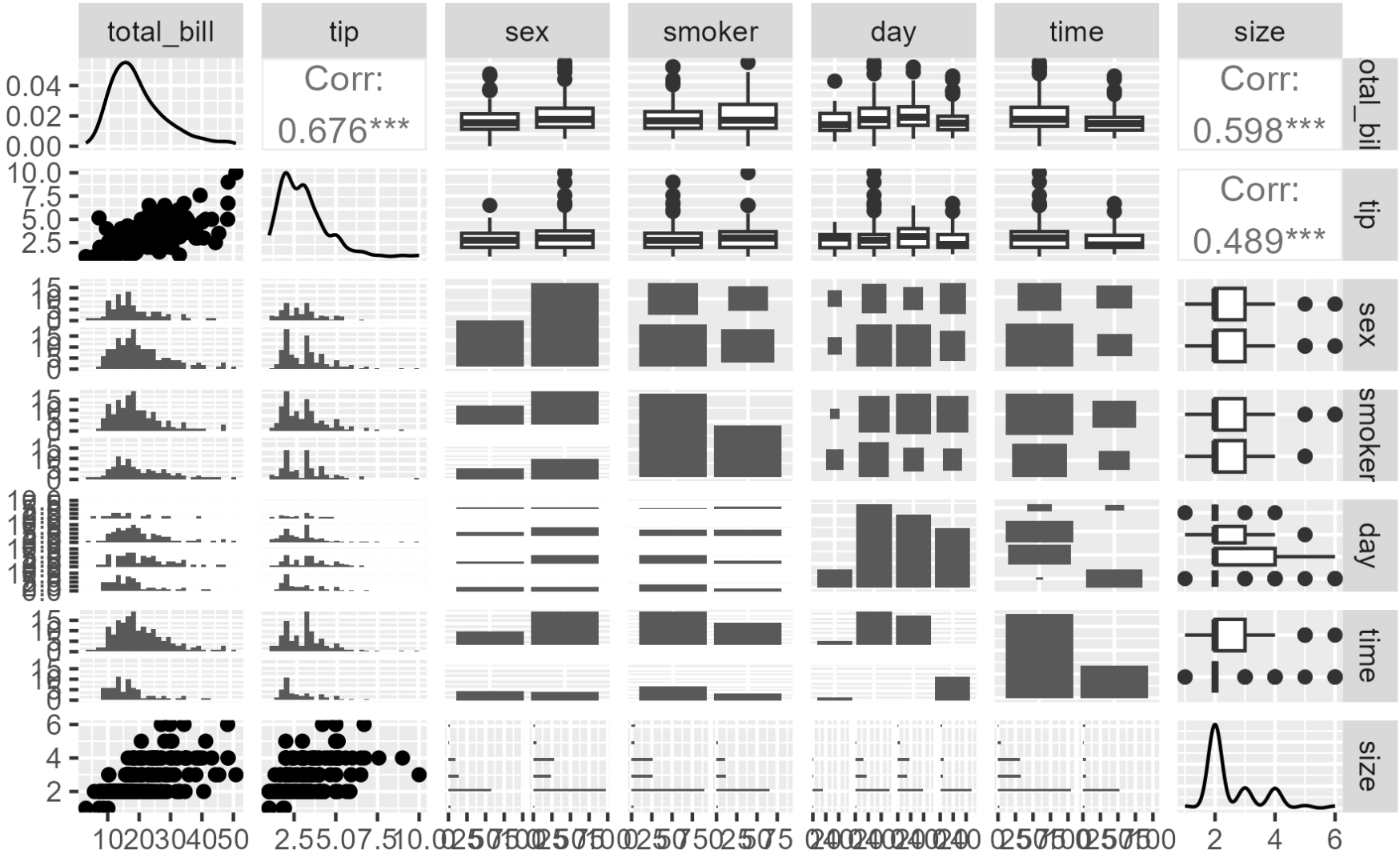
- On the main diagonal we have density plots of the attributes (how frequent are different values of the attribute).
- Below the main diagonal we have the scatterplots. For each scatterplot, the variable on the y-axis is the one whose name is on the corresponding row, while the variable on the x-axis is the one whose name is on the corresponding column.
- Above the main diagonal we have correlations.

- *ggpairs* is a very versatile function, it has many options. The *switch* parameter determines where the line and column labels should be (by default it is right side and bottom), the value *"x"* specifies that the column labels should be at the bottom, *"y"* specifies that the row labels should be at the left side and *"both"* swaps both labels.
- The diagram has three main sections:
  - the main diagonal (*diag*)
  - below the main diagonal (*lower*)
  - above the main diagonal (*upper*).
- The data on the main diagonal can be either *continuous* or *discrete*. The other two sections can be *continuous* (both), *discrete* (both) or *combo* (one continuous and one discrete). For each type of data you can set the type of visualization to use.
- Moreover, you can define aesthetics as well (just like in case of *ggplot*).

- Let's see a few examples of what you can do with the *ggpairs* function, considering a non-time series data set, *tips* (from package *reshape*), containing information recorded by a waiter about received tips. It contains the following attributes:
  - total bill - the total amount to be payed, numeric.
  - tip - the received tip, numeric
  - sex - the sex of the bill payer, discrete
  - smoker - whether there were smokers among the customers, Yes or No
  - day - day of week, discrete (Sunday, Saturday, Thursday, Friday)
  - time - time of the day, discrete (Dinner, or Lunch)
  - size - size of the party, discrete/numeric (numbers 1 to 6)
- **Obs:** Inspiration for these visualizations was taken from `https://ggobi.github.io/ggally/articles/ggpairs.html`

- Let's start by visualizing everything.

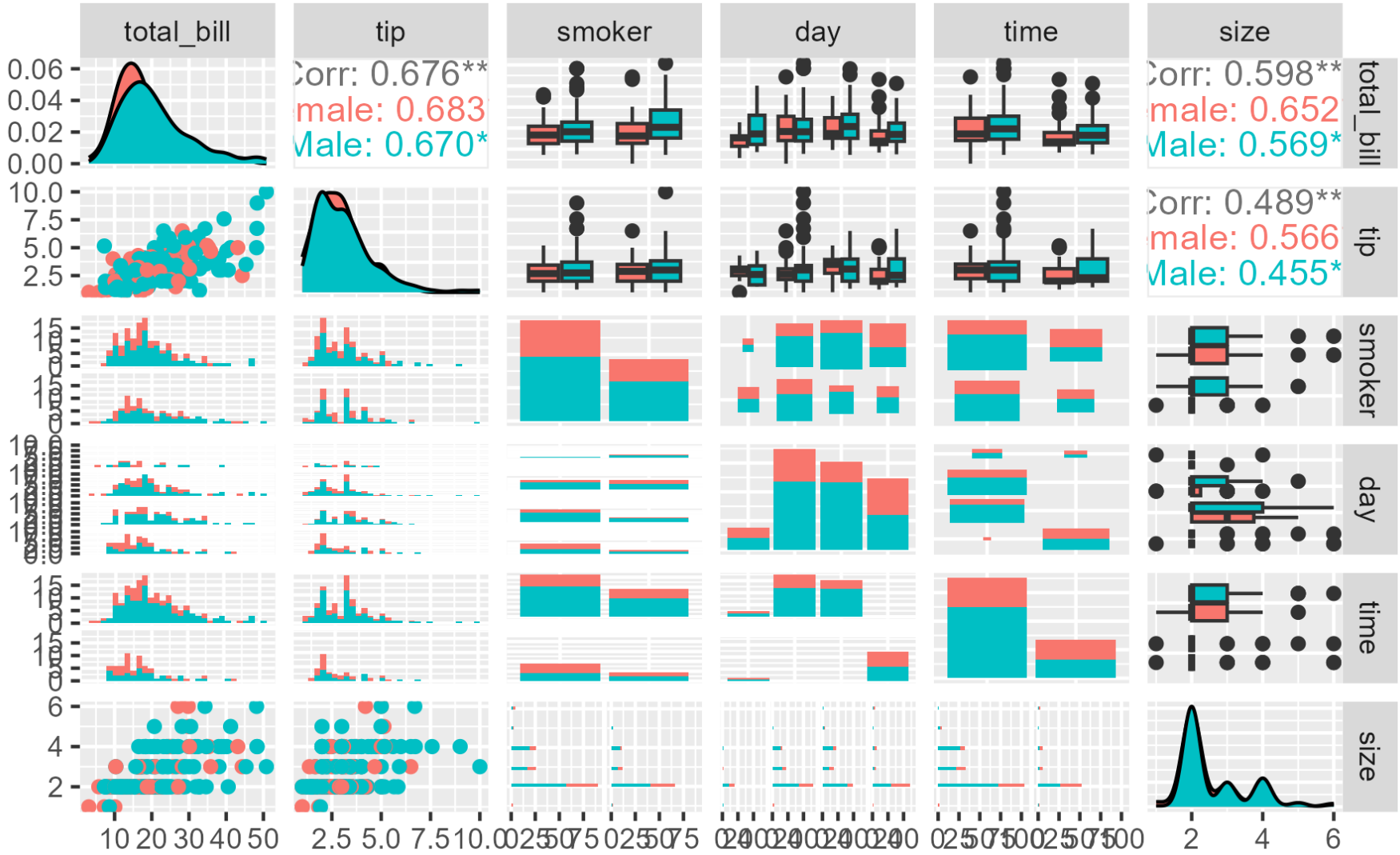
```
ggpairs(tips)
```


total_bill
tip
sex
smoker
day
time
size
Corr:
0.676***
Corr:
0.598***
Corr:
0.489***
total_bill
tip
sex
smoker
day
time
size

- We can see that on the main diagonal we have density plots (for continuous values - total_bill, tip and size) or bar plots with frequencies (the other attributes).
- For the lower part, we have:
  - scatter plot, when both attributes are numerical (combinations of tip, total_bill and size).
  - histogram (facethist), when we have a numerical and a categorical attribute (we have one histogram for every possible categorical value)
  - bar plots (for the combination of possible values) when we have two discrete variables.
- For the upper part we have:
  - boxplots - when we have a continuous and a discrete variable
  - correlation - when we have two continuous variables
  - count plot - when we have two discrete variables (area of the figure is proportional to the count for the combination of values).

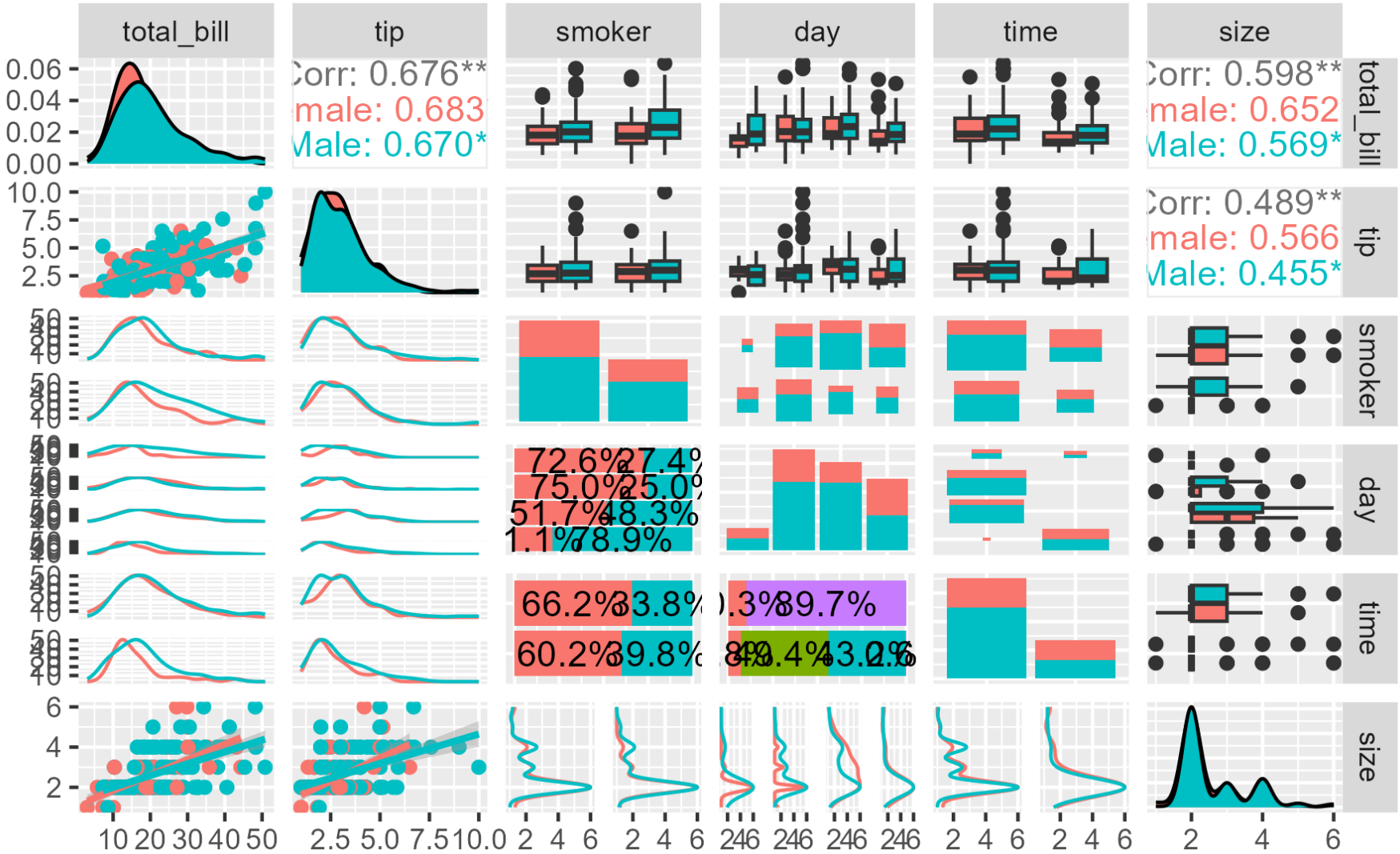
- Let's simplify a little, by removing *sex* and making it an aesthetic.
- We will have the same type of plots, but one attribute less, and everything will be colored according to sex.

```
ggpairs(tips, columns =c(1,2,4,5,6,7), mapping = aes(colour = sex))
```


total_bill
tip
smoker
day
time
size
Corr: 0.676**
male: 0.683
Male: 0.670*
Corr: 0.598**
male: 0.652
Male: 0.569*
Corr: 0.489**
male: 0.566
Male: 0.455*
total_bill
tip
smoker
day
time
size

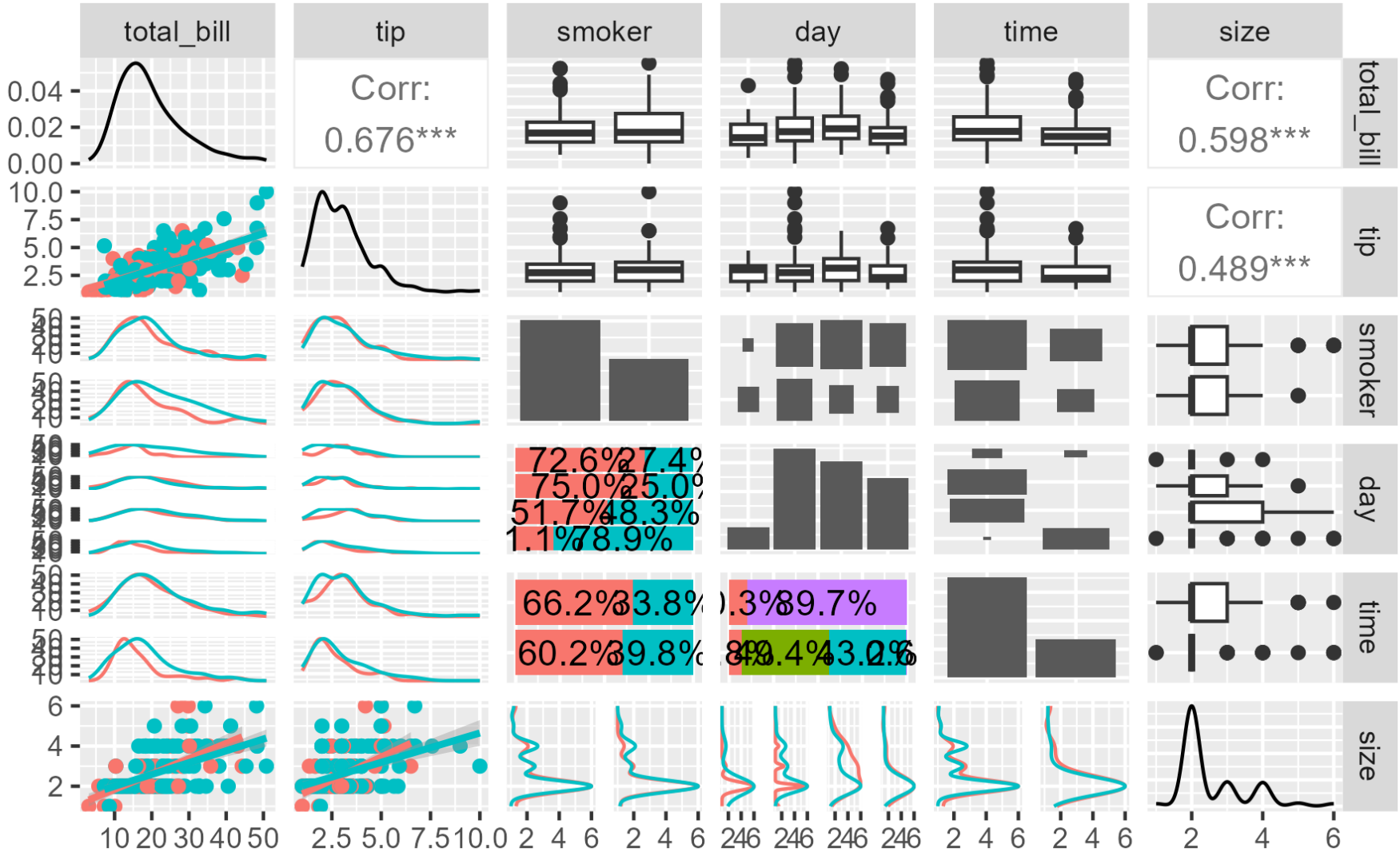
- In order to modify the chart types, we need to use the *lower* and *upper* parameters. They take as value a list, with three name-value pairs.
- The names are *continuous*, *combo*, *discrete*, while the values are the types of the plot to use.
- The complete list of possible plot types for each combination (with examples) can be found here: `https://ggobi.github.io/ggally/articles/ggally_plots.html`
- In the example below we only set custom plot types for the lower part of the matrix. We can similarly set values to the *upper* part as well and to the *diag* part, with the observation that for the diagonal we only have values for *continuous* and *discrete* variables.

```
ggpairs(tips, columns =c(1,2,4,5,6,7), mapping = aes(colour = sex),
    lower = list(continuous = "smooth", combo = "facetdensity", discrete = "
     rowbar"))
```


total_bill
tip
smoker
day
time
size
Corr: 0.676**
Male: 0.683
Male: 0.670*
Corr: 0.598**
Male: 0.652
Male: 0.569*
Corr: 0.489**
Male: 0.566
Male: 0.455*
72.6%
27.4%
75.0%
25.0%
51.7%
48.3%
1.1%
78.9%
66.2%
33.8%
60.2%
39.8%
89.7%
43.2%

- You can add the aesthetic also as part of the upper or lower list, and in this case it will only be applied to the plots on that side.

```
ggpairs(tips, columns =c(1,2,4,5,6,7),
   lower = list(continuous = "smooth", combo = "facetdensity", discrete = "
     rowbar", mapping = aes(color = sex)))
```


total_bill
tip
smoker
day
time
size
Corr:
0.676***
Corr:
0.598***
Corr:
0.489***
total_bill
tip
smoker
day
time
size

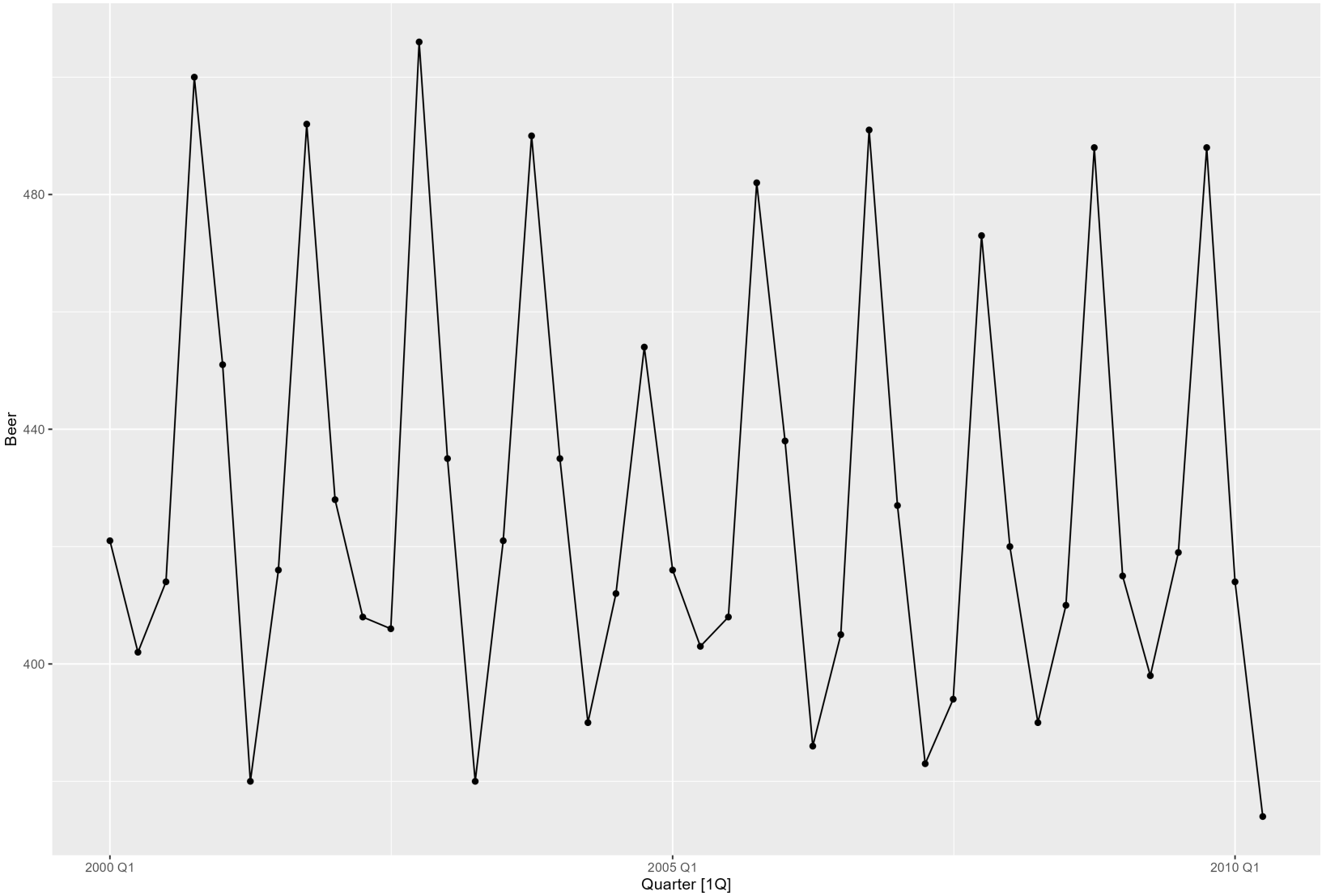
# Lag plots

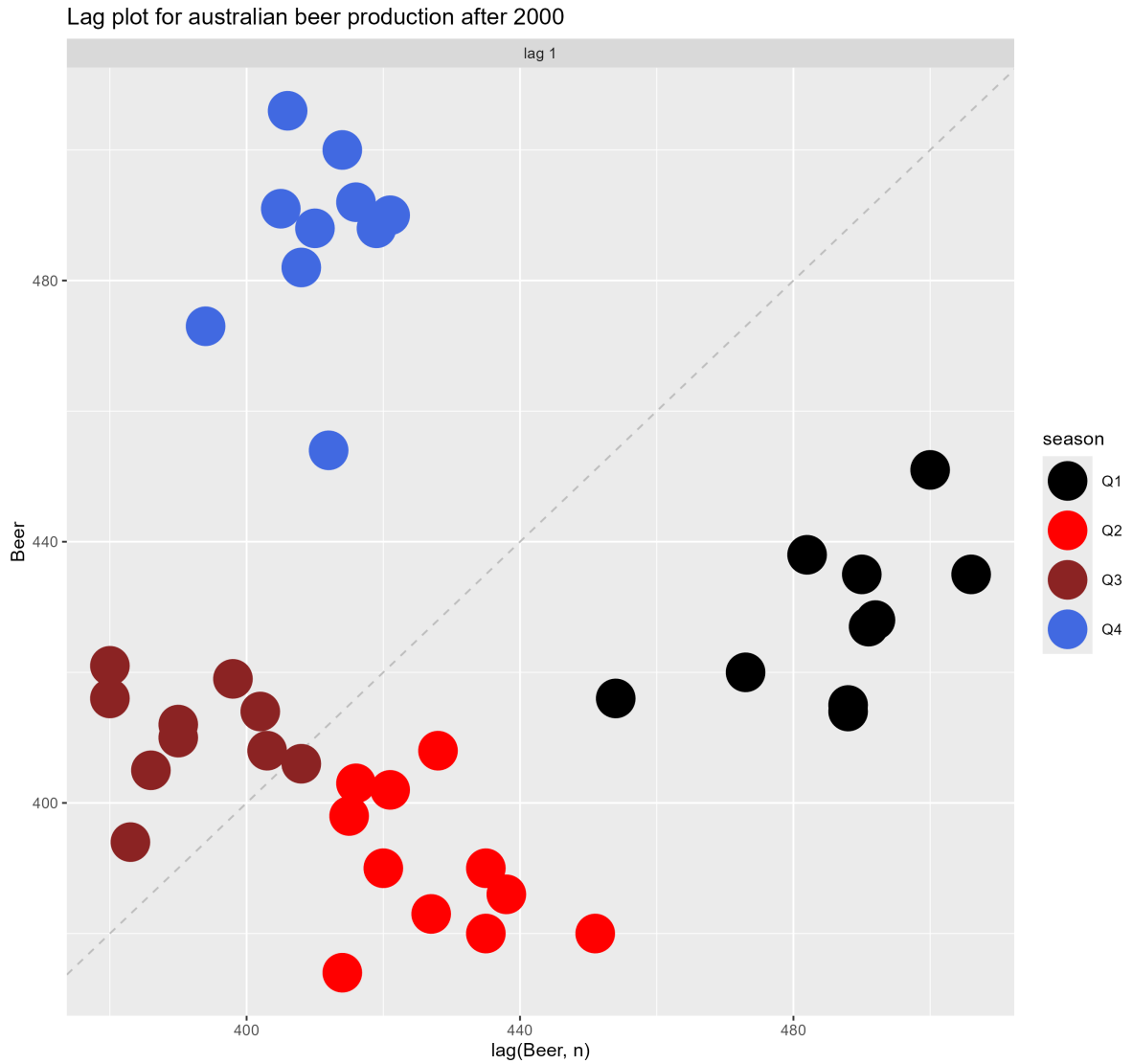
- *Lag plots* are scatter plots, where the time series $y_t$ is plotted against $y_{t-k}$ for different values of k.
- For example, for $k = 1$ we plot the value of $y_t$ against $y_{t-1}$.
- In R a lag plot can be created using the function *gg_lag*

# Lag plots

- Let us consider as example a subset of the *aus_production* data set, which contains beer production after 2000.
- Let us create a scatterplot at a lag = 1.

```
recent_beer <- aus_production |>
  filter(year(Quarter) >= 2000) |>
  select(Quarter, Beer)

recent_beer
recent_beer |> autoplot()

recent_beer |> gg_lag(lags = 1, geom = "point") +
  scale_color_manual(values = c("black", "red", "brown4", "royalblue")) +
  geom_point(size = 10) +
  labs(title = "Lag plot for australian beer production after 2000")
```


480
440
400
Beer
2000 Q1
2005 Q1
2010 Q1
Quarter [1Q]

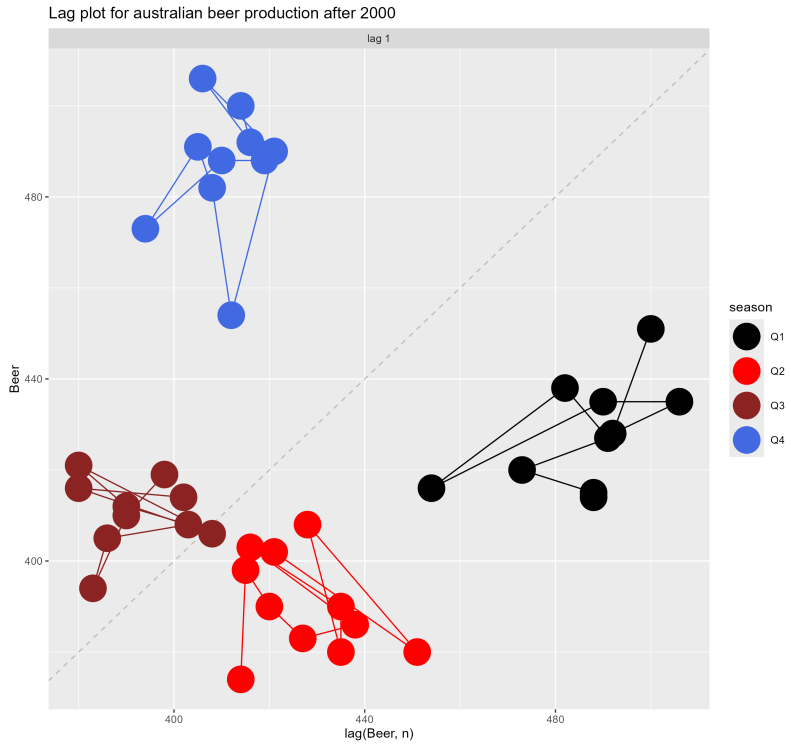

Lag plot for australian beer production after 2000
lag 1
Beer
480
440
400
400
440
480
lag(Beer, n)
season
Q1
Q2
Q3
Q4

# Lag plot for k = 1

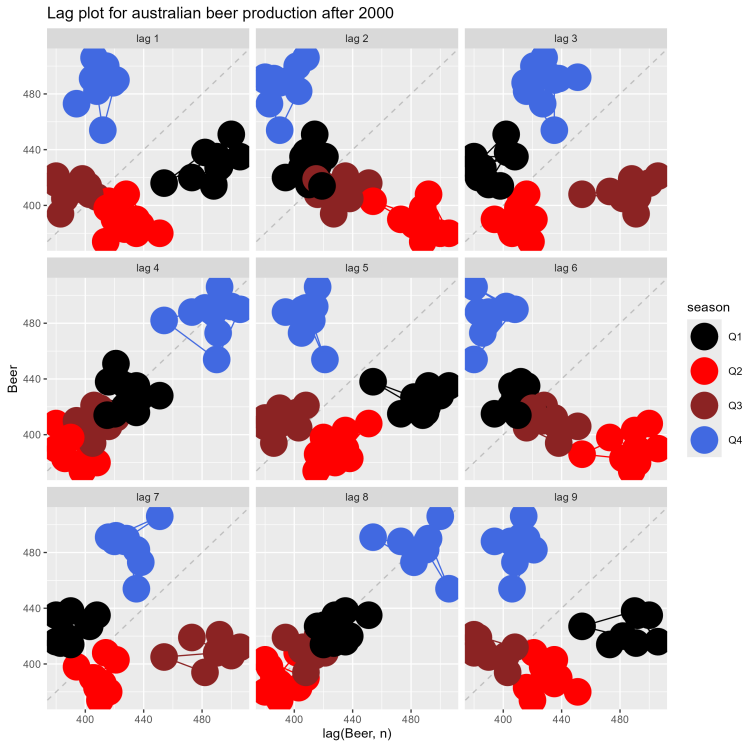
- On the previous slide, we have a plot of $y_t$ (y-axis) over $y_{t-1}$ (x-axis).
- The color of the dots denotes the quarter to which it belongs $\Rightarrow$ the blue dots in the upper left corner correspond to Q4 data, plot against Q3 data (for Q4 at lag 1 you find Q3 data).

# Lag plot for k = 1

- If we remove the *geom* parameter from the *gg_lag* function (or set its value to *path*) points belonging to the same quarter will be connected chronologically.



- If we do not specify the lag for which we want the scatterplot, the *gg_lag* function will display by default plots up until lag 9.

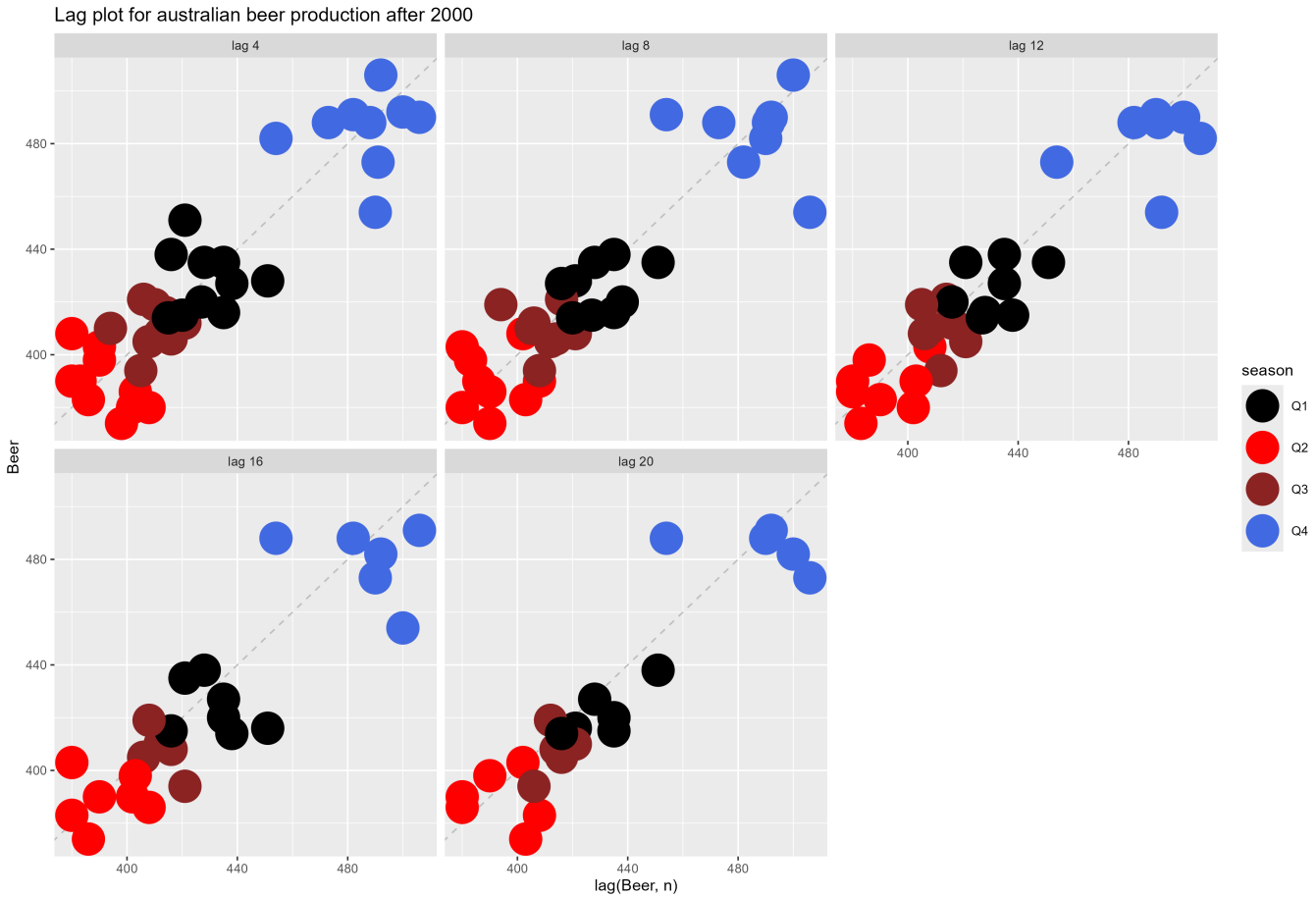



- Is there a lag for which there is a pattern in the data?

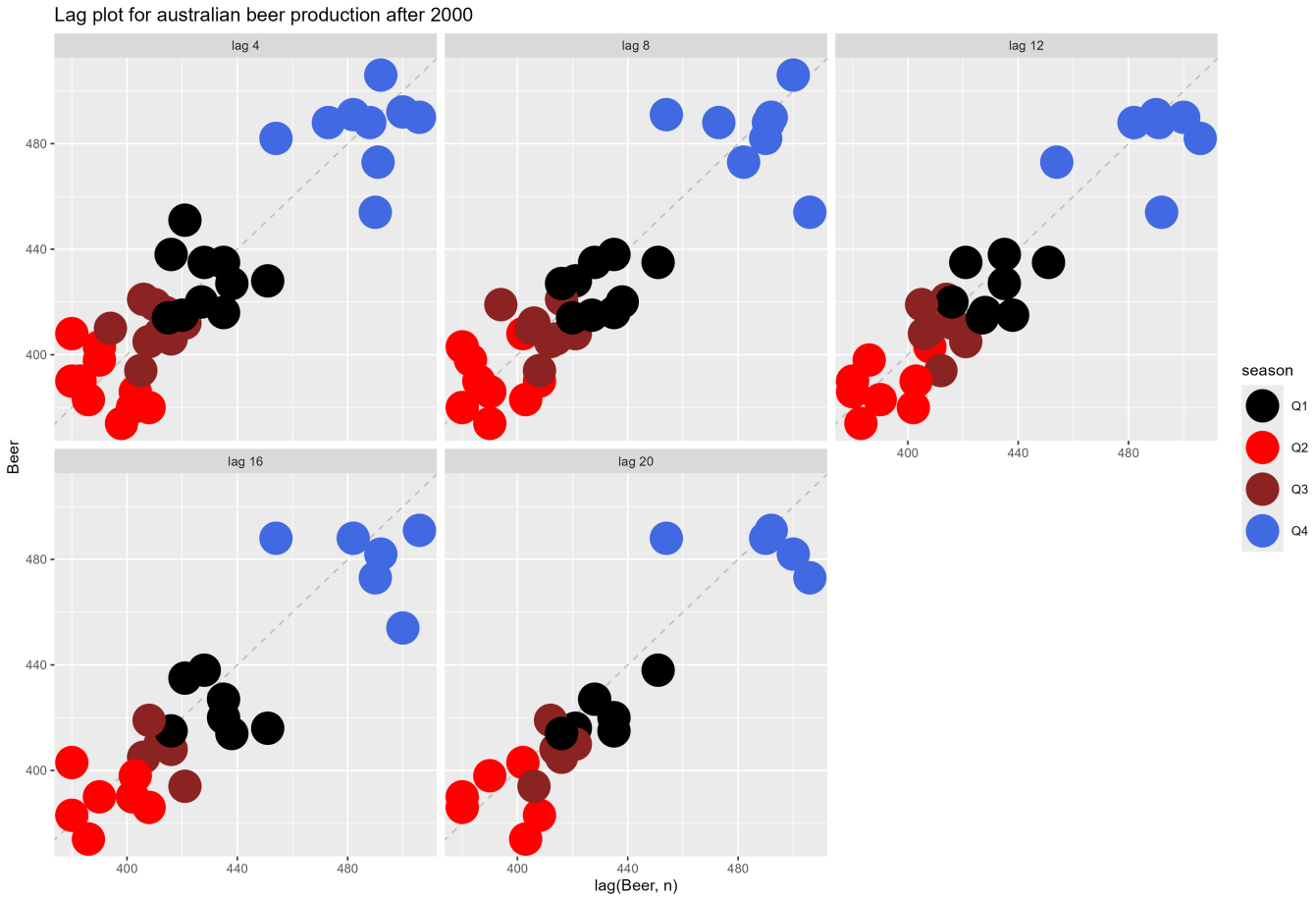
# Selecting specific lags

- You can specify which lags to be used for the plot, using the *lag* parameter, whose value should be a vector (of lags). You can specify intervals (for ex: 2:5) or specific values.

```
recent_beer |> gg_lag(lags = c(4, 8, 12, 16, 20), geom = "point") +
  scale_color_manual(values = c("black", "red", "brown4", "royalblue")) +
  geom_point(size = 4)
```



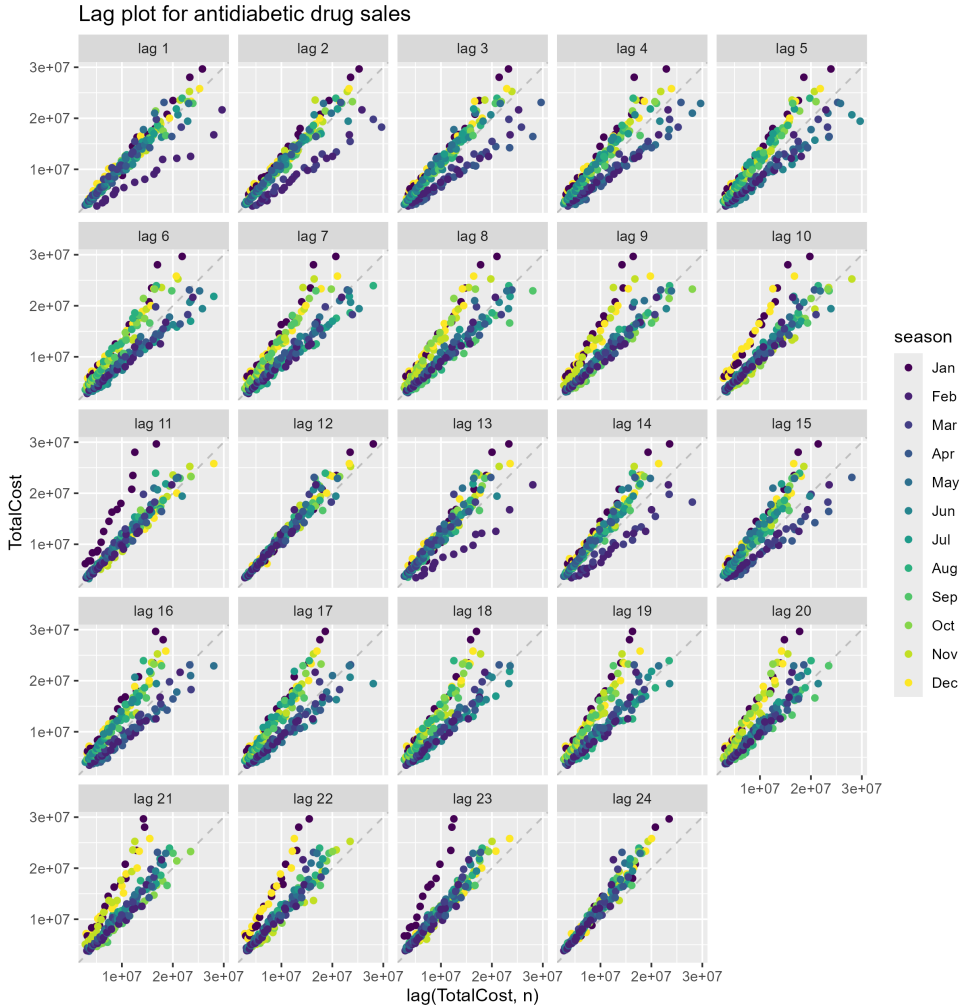

- We can see that there is a strong linear relationship between points and their lagged values at lags which are multiples of 4.
- What do you think is the reason for that?




- We can see that there is a strong linear relationship between points and their lagged values at lags which are multiples of 4.
- What do you think is the reason for that?
- Since data is seasonal, it makes sense to see a relation between points at a lag which is a multiple of 4.

- Let us see the lag plot of the A10 data set (monthly drug prescription for antidiabetic drugs)

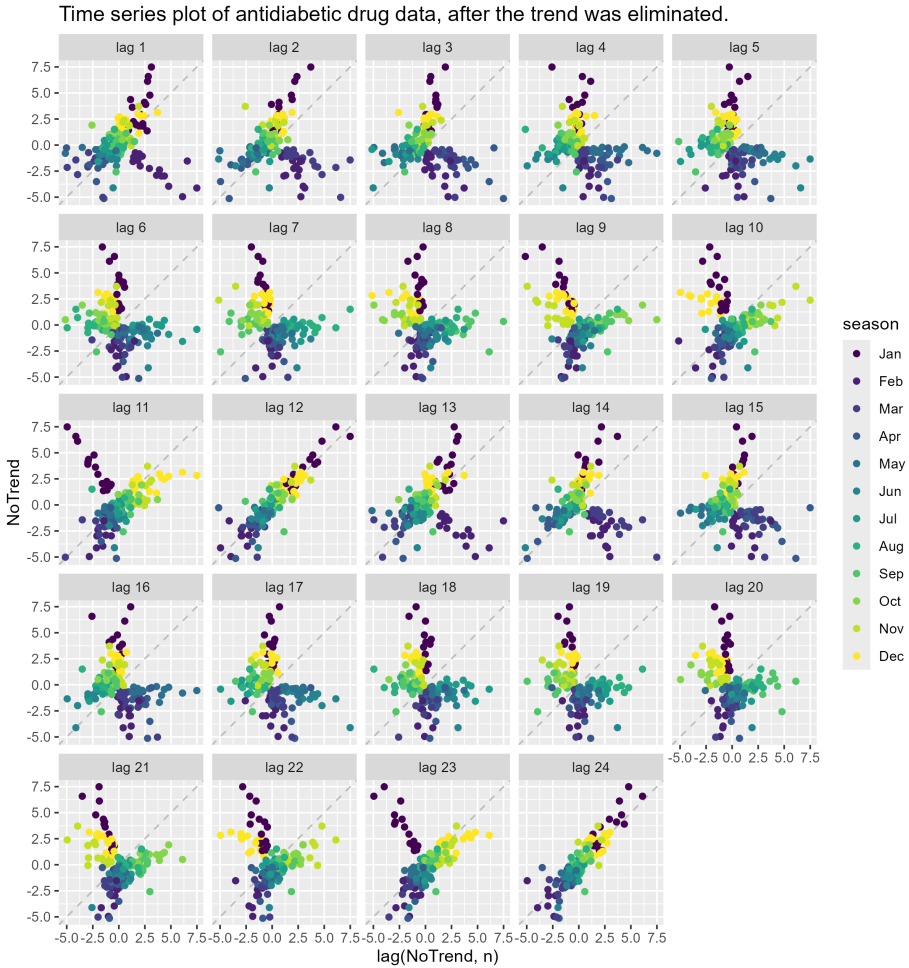
```
a10 |> gg_lag(lags = 1:24, geom = "point") +
  labs(title = "Lag plot for antidiabetic drug sales")
```



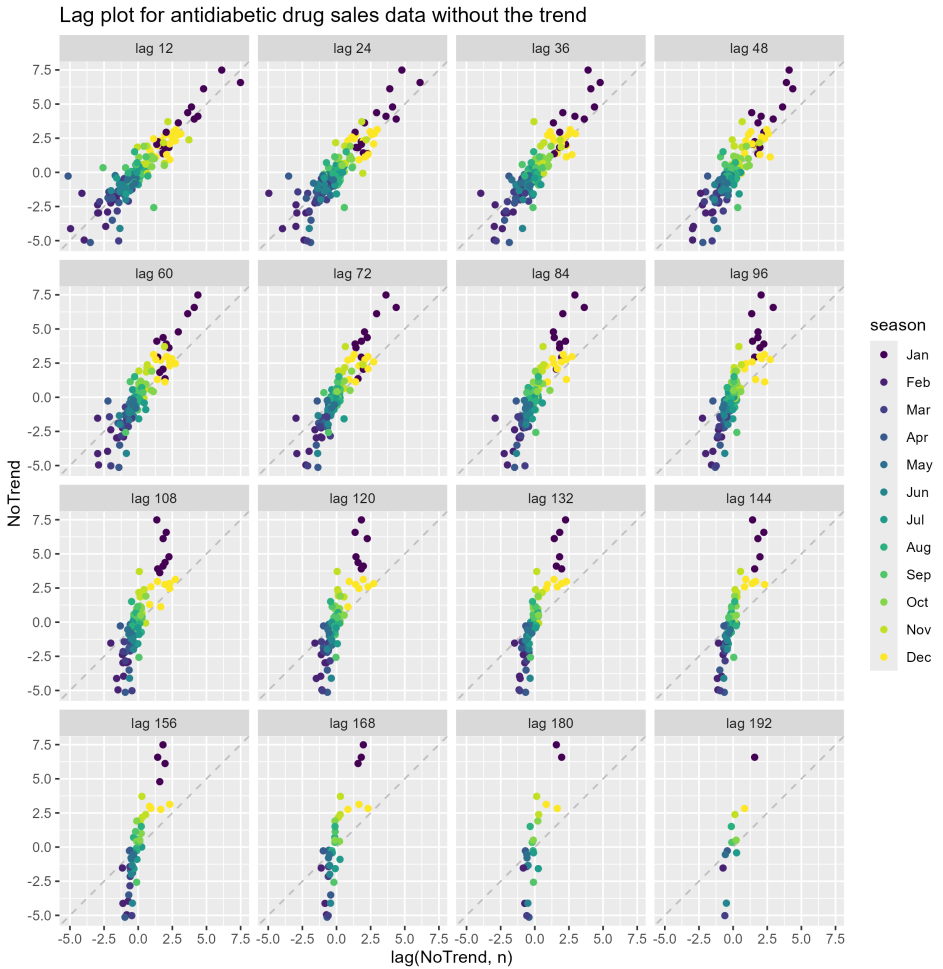

- Why is data correlated at all lags?

- Since the data has a trend, this is visible on the lag plots as well (just like in case of correlations, where trended data can lead to overly inflated correlations).
- Trend can also "hide" actual seasonality.
- If we eliminate the trend, we can see the actual seasonal pattern. (How to eliminate trend will be discussed later)

- After the elimination of trend



- If we only look at lags which are multiples of 12




- Why does the pattern in data change when the lag increases?

- After transforming data to eliminate multiplicative seasonality